

**CHARACTERIZATION OF PROBABILISTIC POLARIZATION IN
FERROELECTRIC PZT THIN FILMS**

by

Rohan S. Pankaj

An Undergraduate Honors Thesis

Submitted to the School of Engineering, Brown University

In partial fulfillment of the requirements for a degree of

Bachelor of Science (Sc.B.) in Electrical Engineering

May 2026

By signing below, I attest that the undergraduate thesis listed above meets the criteria for Honors.

Signed by:

0CD3A7AA1229446...

Advisor: Dr. Lucas Caretta

Assistant Professor of Engineering, Brown University

Signed by:

6C449F9C71A2469...

Reader: Dr. Alexander Zaslavsky

Professor of Engineering and Physics, Brown University

Honors Committee Chair: Dr. Monica Martinez Wilhelmus

Assistant Professor of Engineering, Brown University

© 2026 Rohan S. Pankaj. All rights reserved.

Acknowledgments

There are countless people responsible for helping support the journey that has led to this thesis. First, I want to thank my advisor, Professor Lucas Caretta, for whom I have been working for the better part of two years. As an undergraduate interested in nanotechnology, quantum devices, and precision manufacturing his group has been an incredible place to learn and explore. I have further appreciated Prof. Caretta's mentorship and constant encouragement to pursue my interests and to take full advantage of all the opportunities in the lab. A big thanks also goes to Alexander Qualls, my PhD student mentor, and constant teacher without whom this thesis would certainly not have been possible. Through countless hours of brainstorming and troubleshooting Alex was always willing to help and provided invaluable guidance. I would also like to thank Shih Wei for his help in growing the devices explored in this thesis and for many insightful conversations on PZT and the growth process. George Fratain also deserves special thanks for his mentorship and collaboration on developing the measurement set up used in this thesis. I also extend deep appreciation for Hector Garces for teaching me to use tools such as the AFM, PFM, and XRD as well as the rest of the Caretta Lab for being so welcoming over these past few years.

Professor Alexander Zaslavsky also deserves special thanks for serving as my academic advisor and the second reader of this thesis. It was through his courses in E&M and semiconductor devices that I first got exposure to this field. I would further like to thank my past PIs and mentors including Prof. Greg Landsberg, Prof. Loukas Gouskos, Dr. Marko Stamenkovic, and Prof. Carmille Palmer.

A great deal of thanks is also owed to the several friends I have made here at Brown. There are too many to name, but to Bering Edwards, Justin Meszler, Carson Bauer, Margaret Lorraine, Ella Hochstadt, Elizabeth Gresfali, Ian Hajra, Ethan Cooperman, Wyatt Sieminski, Joshua Ginzburg, Glenn Wang, Olivia Cox, and Justin Kingsnorth, I count you all dear friends and cannot imagine a college experience without you. I would like to also thank Hayden Miller, Gavin Pitt, Jason Huang, and Jordan Pfeifer, my CERN colleagues, for being an incredible cohort.

Finally, and most importantly, I would like to thank my family. To my parents, Prof. Pankaj Jaiswal and Prof. Sushma Naithani, thank you for your support and guidance. To my brother Ravi Pankaj, thanks for being a constant source of fun and a reminder to not take things so seriously. Lastly, thank you to my grandparents JP Naithani, Godavari Naithani, Rakesh Jaiswal, and particularly MC Jaiswal, my grandfather and electrical engineer, with whom building robots as a kid inspired me to pursue electrical engineering.

Sincerely,

Rohan S. Pankaj

CHARACTERIZATION OF PROBABILISTIC POLARIZATION IN FERROELECTRIC PZT THIN FILMS

by

Rohan S. Pankaj

Submitted to the Brown University School of Engineering
in April 2026 in partial fulfillment of the requirements of the degree of
Bachelor of Science (Sc.B.) in Electrical Engineering

Abstract

Materials based approaches to True Random Number Generation (TRNG) are an active area of research with a wide range of applications in modeling, simulation, and cybersecurity. We present the detailed theory and design of a FeCAP which utilizes inherent stochasticity in ferroelectric switching dynamics to create a Physically Unclonable Function (PUF). This is done by using an oxygen-deficient YBCO layer as the top electrode in the capacitor. In doing so, we aim to engineer oxygen vacancies at the PZT/YBCO interface to act as pinning sites for the propagation of polarization switching. This would result in a Barkhausen-like effect where in stochastic current spikes can be measured and form the basis of a probability distribution. We present data showing cycle-to-cycle variation in switching dynamics well above circuit noise. These measurements resemble the intended Barkhausen-like behavior. This variation was ultimately attributed to incomplete electrical contact between measurement probes and the rough YBCO layer. Still, the wide distribution created by changes in switching dynamics signal that the proposed approach remains an enticing avenue for TRNG.

Contents

INTRODUCTION.....	8
1.1 MOTIVATIONS.....	8
1.2 A PRIMER ON FERROELECTRICS.....	9
1.3 THE FERROELECTRIC CAPACITOR (FeCAP).....	10
1.4 POLARIZATION SWITCHING	11
1.5 SUMMARY OF WORK	12
LITERATURE REVIEW.....	13
2.1 STOCHASTIC SWITCHING AND DOMAIN DYNAMICS	13
2.2 FERROELECTRIC APPROACHES TO TRNG	13
2.3 THE SWITCHING DYNAMICS APPROACH	14
2.4 THE MATERIAL STACK	14
METHODOLOGY	16
3.1 SAMPLE GROWTH	16
3.2 HYSTERESIS MEASUREMENTS.....	16
3.3 PUND MEASUREMENTS	19
3.4 MEASUREMENT SET UP	21
3.5 DATA ANALYSIS	21
3.6 AFM/PFM.....	22
3.7 XRD	22
RESULTS.....	24
4.1 TYPICAL SWITCHING BEHAVIOR IN FeCAPs.....	24
4.2 INTERRUPTED SWITCHING BEHAVIOR IN STOCHASTIC FeCAPs.....	26
4.3 DISTRIBUTION OF PUND VALUES	28
4.4 MATERIALS AND GROWTH VALIDATION	31
DISCUSSION AND CONCLUSION	35
5.1 INTERPRETATION OF CHARGE TRANSPORT MEASUREMENTS.....	35
5.2 CREATING A PUF FROM PUND DATA	36
5.3 OTHER SOURCES OF STOCHASTICITY	36
5.4 LIMITATIONS	37
5.5 CONCLUSION AND FUTURE WORK	38
REFERENCES.....	39

List of Figures

FIGURE 1 DOUBLE WELL POTENTIAL MODEL OF THE FERROELECTRIC POLARIZATION	9
FIGURE 2 ILLUSTRATION OF DOUBLE WELL FERROELECTRIC SWITCHING	10
FIGURE 3 TRADITIONAL CAPACITOR (LEFT) V. A FECAP (RIGHT).....	11
FIGURE 4 A DIAGRAM OF DOMAIN-WALL PROPAGATION PINNED BY LOCAL DEFECT	12
FIGURE 5 MATERIAL STACK OF STOCHASTIC FECAPS	15
FIGURE 6 ILLUSTRATION OF A HYSTERESIS MEASUREMENT	17
FIGURE 7 FEATURE EXTRACTION FROM FERROELECTRIC HYSTERESIS	18
FIGURE 8 THE EFFECTS OF VARIOUS NONIDEALITIES ON HYSTERESIS LOOPS.....	19
FIGURE 9 BREAKDOWN OF PUND MEASUREMENTS	20
FIGURE 10 MEASUREMENT CIRCUIT	21
FIGURE 11 AFM SET UP	22
FIGURE 12 AN ILLUSTRATION OF BRAGG'S LAW.....	23
FIGURE 13 XRD SET UP	23
FIGURE 14 SIMULATION OF MATERIAL STACK USING LGD	24
FIGURE 15 CONTROL HYSTERESIS ON SRO/PZT/PT USING THE MEASUREMENT SET UP	25
FIGURE 16 CONTROL PUND ON SRO/PZT/PT USING THE MEASUREMENT SET UP	26
FIGURE 17 STOCHASTIC HYSTERESIS EXAMPLES ACROSS GROWTHS AND DEVICE SIZE.....	27
FIGURE 18 EXAMPLE HYSTERESIS LOOPS FOR A 10x10 μm^2 DEVICE ON SRO/PZT/YBCO-2	27
FIGURE 19 STOCHASTIC PUND EXAMPLES ACROSS GROWTHS AND DEVICE SIZE	28
FIGURE 20 P^* V. CYCLE FOR A. SRO/PZT/YBCO-1 AND B. SRO/PZT/YBCO-2.....	29
FIGURE 21 DISTRIBUTION OF P^* VALUES FOR A 40x40 μm^2 FECAP ON SRO/PZT/YBCO-1.	30
FIGURE 22 BREAKDOWN OF INDIVIDUAL PUND DISTRIBUTIONS FOR SRO/PZT/YBCO-1	31
FIGURE 23 AFM OF UNPATTERNED SRO/PZT/YBCO-3	32
FIGURE 24 AFM OF PATTERNED DEVICE ON SRO/PZT/YBCO-3	32
FIGURE 25 AFM OF PATTERNED DEVICE ON SRO/PZT/YBCO/Pt-3.....	33
FIGURE 26 CHARGE TRANSPORT DATA TAKEN ON SRO/PZT/YBCO/Pt-3	33
FIGURE 27 XRD OF SRO/PZT/YBCO-3 WITH LABELED PEAKS	34

List of Tables

TABLE 1 GROWTH PARAMETERS FOR STOCHASTIC SAMPLES..... 16

Chapter 1

Introduction

1.1 Motivations

The field of random number generation (RNG) is an active area of research with important applications in fields like modeling, simulation, and cybersecurity [1]. For the latter, RNG is used in important protocols such as the generation of session keys, digital signatures, masks, and challenges for authentication [1]. For these cybersecurity applications, it is incredibly important that RNG is unpredictable. There are two main types of random number generation. The first is True Random Number Generation (TRNG), with the second being Pseudo-Random Number Generation (PRNG) [1]. PRNG uses “deterministic mathematical algorithms” to generate sequences of numbers that are only random in appearance [1]. Because of this, PRNG sequences can be predicted by knowing the state of the system. Thus, TRNG is preferred in cryptography.

True random number generation comes from hardware devices that extract randomness from their operation [1]. This randomness often comes from a combination of manufacturing imperfections, thermal noise, and interference, giving each device a unique distribution “fingerprint” that cannot be replicated. Such devices are said to have Physically Unclonable Functions (PUFs) and are considered the most secure in cryptographic protocols. The most common PUF methods today are based on imperfections in CMOS circuits [1]. However, these methods have shown some vulnerability to machine learning based attacks [2]. This has prompted increased scholarship into PUFs based on entropy sources with inherent physical unpredictability [3, 4].

Ferroelectric (FE) thin films have long been studied for promising applications in low-power memory and computing devices. These devices operate by switching polarization states in the presence of an applied field. For such applications, these switches are designed to be deterministic and repeatable. We investigate a ferroelectric device where switching appears to be stochastic. The device in question is a ferroelectric capacitor where lead zirconate titanate (PZT) is sandwiched between a strontium ruthenate (SRO) bottom electrode and a yttrium barium copper oxide (YBCO) top electrode. In this thesis we explore the potential causes for stochasticity in ferroelectric polarization shown in this material stack and discuss the implications for future devices.

1.2 A Primer on Ferroelectrics

Ferroelectrics are a special type of dielectric that exhibit two characteristic behaviors [5]. First, they have at least two orientational states in the absence of an electric field. And second, applying an electric field can shift the materials from one state to another. The crystal structure of these two orientational states is identical, except for the direction of electric polarization vectors which give rise to the two states.

Thus, ferroelectricity is a property that is highly associated with specific crystalline structures. Of the 32 crystal classes, 20 exhibit electrical polarity in the presence of stress [5]. Conversely, they also exhibit stress when an electric field is applied. Materials with these crystalline properties are called piezoelectric. Ten of these crystal classes have unique polar axis, meaning they have an electric moment per unit volume, they are referred to as the pyroelectric classes. Ferroelectrics are a subset of these pyroelectric classes which maintain polarization in the absence of applied fields.

An intuitive way to understand ferroelectric polarization switching comes from the double well model, as seen in Figure 1. Each well represents an equilibrium polarization state. The barrier between them is the potential difference required to switch polarization states.

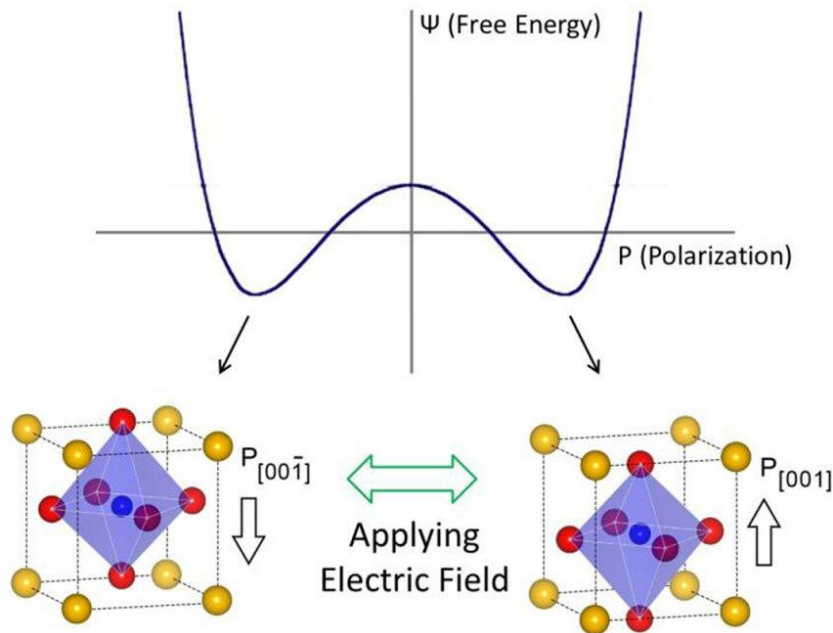


Figure 1 Double well potential model of the ferroelectric polarization: Here, the polarization (P) of ferroelectric $\text{PbZr}_{0.2}\text{Ti}_{0.8}\text{O}_3$ is produced by displacive distortions of the cations and oxygen octahedral in the crystal lattice. The ferroelectric can adopt one of two degenerate polarizations ($P_{[001]}$ and $P_{[00\bar{1}]}$) along the $[0\ 0\ 1]$ axis of tetragonal unit cell to minimize its free energy. In principle, applying an electric field larger than a critical value can overcome the potential barrier and lead to ferroelectric switching. Image and caption reprinted from [6].

By applying an electric field, we can tilt the well and thus switch polarization states. An example of this is shown in Figure 2.

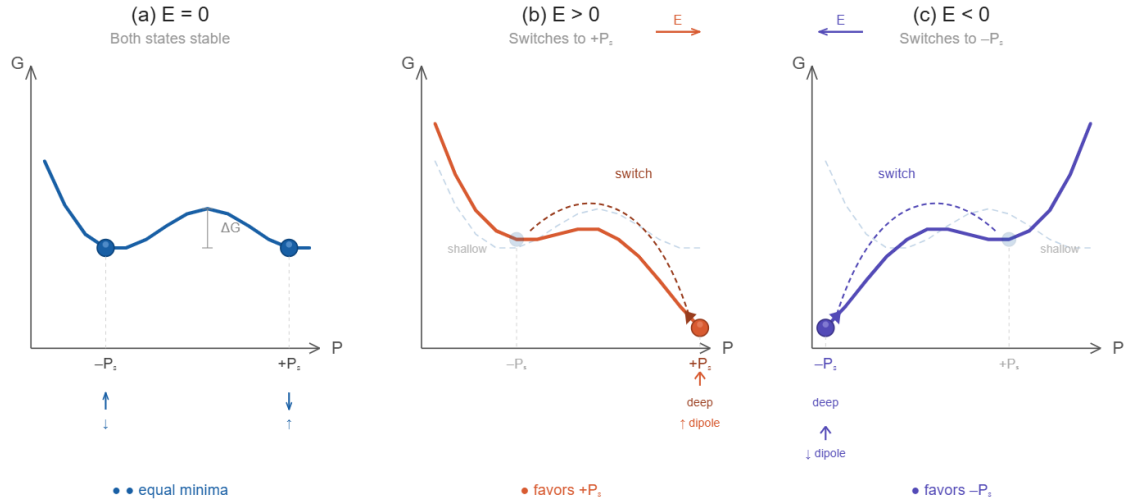


Figure 2 Illustration of Double Well Ferroelectric Switching:

We see in **a.** when there is no applied field, both polarization states are equally favorable. **b.** when a positive electric field is applied the potential well tilts and the positive polarization state become more favorable. **c.** when a negative electric field is applied the potential tilts and the negative polarization state becomes more favorable.

The double well model also allows us to simulate ferroelectric switching behavior using the Landau-Ginzburg-Devonshire (LGD) theorem [7]. The LGD theorem models the free energy density ($\mathcal{F}_P$) in the system as:

$$\mathcal{F}_P = \frac{1}{2}aP^2 + \frac{1}{4}bP^4 + \frac{1}{6}cP^6 - EP \quad (1)$$

Where E is the applied electric field and P is polarization. A relation between the applied electric field and polarization state can then be found by taking the minima of $\mathcal{F}_P$.

$$E = aP + bP^3 + cP^5 \quad (2)$$

The coefficients a , b , and c are referred to as Landau coefficients, and are unique to different ferroelectric materials. Further adjustments to the LGD theorem need to be made for the thin film case, where epitaxial strain between interfaces also plays a role [8]. This is discussed in detail in the capstone paper by Pankaj [9], and for this thesis we will utilize the accompanying opensource software FeSim to model devices.

1.3 The Ferroelectric Capacitor (FeCAP)

To induce and measure the switching behavior shown in Figure 2 an electric field needs to be applied parallel to the polarization direction of the ferroelectric. A common way to do this is with a ferroelectric capacitor or FeCAP. In FeCAPs a ferroelectric material replaces the dielectric used in a traditional capacitor. Applying a field to electrodes on

either side of the ferroelectric triggers the switching mechanism. Thus, FeCAPs are an effective way to measure antiparallel ferroelectric polarization switching.

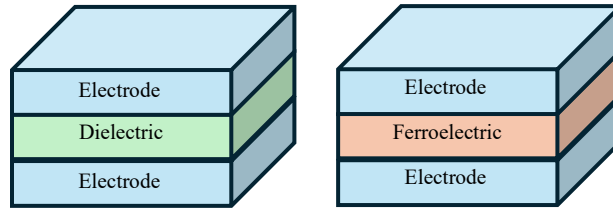


Figure 3 Traditional Capacitor (left) v. a FeCAP (right)

Because FeCAPs maintain a polarization state in the absence of an applied electric field, they are an attractive option for memory storage devices. Expectedly, this application has driven much of the excitement in ferroelectric materials over recent decades leading to products such as Ferroelectric RAM (FeRAM) [10]. However, the two discrete states present in an FeCAPs also make it an interesting candidate for a stochastic device. The introduction of material defects and thermal noise could provide probabilistic switching between the two polarization states creating a PUF that can be used in TRNG.

1.4 Polarization Switching

Ferroelectric switching dynamics have been studied for decades [5, 6, 11-17] and are well understood. Merz and colleagues began this process by identifying that ferroelectric switching begins with several random nucleation events [14]. Later scholarship showed that these nucleation sites propagate outward spreading throughout the ferroelectric [11, 13, 15] as expanding domain walls.

For several years, this intuitive understanding led to the belief that switching polarization was a linear process in time. However, more recent scholarship has shown that this is not the case [12, 16]. Instead, domain growth is a creep process where local pinning potentials modify the double well leading to starting and stopping of domain wall motion. Defects are a common source of these pinning sites and can lead to fractalization of domain growth [12, 16]. One potential source of these defects are oxygen vacancies, which have been shown to restrict ferroelectric domain switching [17]. An example from Jo et al is shown in Figure 4.

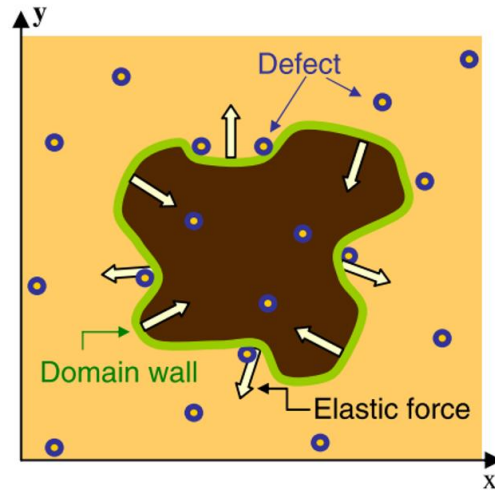


Figure 4 A diagram of domain-wall propagation pinned by local defect: We can see that domain wall motion stops at defect sites. During FE switching overcoming these switching barriers occur when sufficient field has is achieved. Reprinted from [16].

When sufficient field is present to overcome domain wall pinning, the rapid polarization reversal can lead to current spikes when measuring switching dynamics. These are referred to as Barkhausen pulses [5].

1.5 Summary of Work

In this thesis we explore the use of switching dynamics in ferroelectric capacitors to generate random numbers. Recent work, summarized later, has explored the use of ferroelectric devices to generate random numbers. However, these approaches have largely looked at the end state of polarization switching, rather than the actual switching dynamics. This makes it difficult to decouple device level stochasticity from general circuit noise. In this thesis we discuss stochastic-like switching behavior in a FeCAP material stack and investigate the causes.

Chapter two will discuss prior literature on stochastic switching, ferroelectric approaches to TRNG, and a justification for the material stack used in this thesis. Chapter three will follow with a discussion of the methods. This will include detailed descriptions and justifications for the charge transport measurements used later as well as background on the material growth and characterization. Chapter four will present results to be discussed in Chapter 5.

Chapter 2

Literature Review

2.1 Stochastic Switching and Domain Dynamics

The nucleation of polarization switching domains in ferroelectrics are random, and vary from cycle to cycle [13]. Thus, the defects with which growing domains interact also vary from cycle to cycle. As discussed in Chapter 1.4, polarization switching is a creep process, influenced by local defects. The random nucleation coupled with the influence of defects therefore means that switching dynamics also vary from cycle to cycle. In this thesis we explore the possibility of exploiting these variances to create a PUF.

In thin film devices, the interface between the ferroelectric and electrode layer can play a significant role in influencing these dynamics. As found by Han and colleagues [17], the presence of oxygen vacancies at the electrode interface led to the pinning of propagating domains, and even nonswitchable domains. We can thus expect that if defects are spread unevenly throughout the interface, different regions of nucleation will have different volumes of ferroelectric polarization that can be switched. This variation could serve as the basis for a PUF. The influence of defects is further studied in [6]. Further variation comes from thermal noise, which also plays a role in ferroelectric switching [16].

As mentioned earlier, when propagating domains overcome pinning, after sufficient switching field has been built up, there can be a rapid change in polarization volume. This sudden process leads to a spike in current across the electrodes in an FeCAP. These are referred to as Barkhausen pulses which have particularly been shown in PZT, the ferroelectric material focused on in this thesis [18].

2.2 Ferroelectric Approaches to TRNG

The inherent randomness of ferroelectric polarization switching has already gathered some interest in applications in true random number generation. Mulaosmanovic and colleagues [19] were the first to implement this idea. Their approach involved using a pulse sequence near the coercive field that switched a FeFET between its two polarization states generating random bits (1s and 0s) with two near equally probable states.

Later work by Chatterjee [4] and Chien [3] expanded upon this work with FeFETs and FeCAPs respectively. Both were able to maintain a nearly 50% switching probability between two states while also resisting GAN, LSTM, and Fourier regression attacks that have plagued TRNG approaches in the past. Devices proposed by both sets of researchers were shown to pass NIST SP 800-22 standards for TRNG [20]. Li et al [21] further

tackles the practical implementations of such a device by addressing power and reconfigurability challenges.

While the approaches mentioned above have largely involved the use of HFO in a FeFET/FeCAP set up, some recent scholarship has also generalized the approach to LiNbO₃ [22] and 2D Ferroelectric Tunnel Junctions (FTJs) [23].

In these prior approaches, a switching pulse is applied to the ferroelectric, and the resulting state of the device is read through drain current, resistance, or threshold voltage. If these measurements cross a predetermined threshold, the output is recorded as a '1' or otherwise, a '0'. By repeatedly applying the switching pulse and reading the output these devices generate a constant bitstream of 1s and 0s.

A formal justification on the suitability of ferroelectrics for TRNG applications is further provided by Ramesh and colleagues [24].

2.3 The Switching Dynamics Approach

Through the approaches discussed above, ferroelectrics have demonstrated promising potential for TRNG applications. However, they all rely on reading the end polarization state after a switching pulse is applied. This could make devices vulnerable to the effects of noise, drift, and aging. We explore a novel approach to ferroelectric TRNG that extracts stochasticity from the switching trace itself. Thus, we hope to examine the stochastic role of creep, defects, and Barkhausen pulses directly.

2.4 The Material Stack

Strontium titanate (STO) was selected as the substrate for the device as a standard for the epitaxial growth of perovskites. Strontium ruthenate (SRO) was then grown on the substrate material to serve as the bottom electrode. SRO was selected for its metallic perovskite oxide properties and its good lattice match with the STO substrate and ferroelectric PZT layer above.

Lead zirconate titanate (PZT) was chosen as ferroelectric layer for several reasons. First and foremost, it is a well-studied ferroelectric with high remnant polarization. Second, prior literature has documented Barkhausen switching behavior [18] in thin film layers of PZT, along with the stochastic creep process that we hope to exploit [12, 16].

In designing the FeCAP, we aim to introduce oxygen vacancies to amplify the effects of stochastic nucleation, creep, and domain wall motion in the ferroelectric layer. Careful selection of the top electrode layer was thus needed in order to control the introduction of defects and potentially investigate them with microscopy [25]. Yttrium barium copper oxide (YBa₂Cu₃O_{7-x} or YBCO) was chosen exactly for this purpose. YBCO is a conductive perovskite oxide that has shown epitaxial compatibility with PZT [26]. In their work, Ramesh and colleagues observed that the use of YBCO electrodes with PZT

reduced ferroelectric fatigue [26]. This is an important consideration for TRNG devices as the PUF should not change with cycling of the device. YBCO is also inherently grown oxygen-deficient, so the interface hosts oxygen vacancies that can act as domain wall pinning sites. Furthermore, there has been promising recent scholarship on engineering oxygen vacancy distributions at material interfaces [27]. Such oxygen vacancies are highly tied to influencing domain wall pinning, and motion [27, 28]. The final material stack is shown in Figure 5.

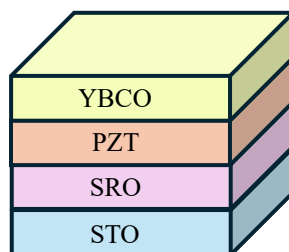


Figure 5 Material Stack of Stochastic FeCAPs

Chapter 3

Methodology

3.1 Sample Growth

The samples were grown via Pulsed Laser Deposition, wherein a laser is fired at a target, creating a plasma which deposits thin layers of atoms on the substrate. In this case an STO substrate was used with SRO grown on top before PZT and YBCO were deposited. PLD was chosen due to the precise control it offers in thin film oxide deposition [29]. Furthermore, it has been shown to produce high quality epitaxial growth of YBCO on PZT [26]. The growth parameters used for the devices discussed in this thesis are shown in Table 1. Growth was done with the aim of creating a 25 nm SRO layer, 50 nm PZT layer, and 40 nm YBCO layer based on previous development samples.

Table 1 Growth Parameters for Stochastic Samples

Sample	Layer	O ₂ Pressure (Torr)	Laser Energy (KV)	Dep Time (min)	Laser Freq. (Hz)	Temp Setpoint (*C)	Anneal Time (min)	Anneal Temp (*C)	Chamber Pressure (Torr)	Energy (mJ)
SRO/PZT /YBCO-1	PZT	0.125	19	25	10	950&940	30	550	760	60
	YBCO	0.125	18	4	10	950	30	600	200	99
SRO/PZT /YBCO-2	PZT	0.125	18.3	10	10	950	-	-	-	90
	YBCO	0.125	18.3	2.5	10	950	-	-	-	99
SRO/PZT /YBCO-3	PZT	0.125	18.6	20	10	885	-	-	-	90
	YBCO	0.125	18.6	2.5	10	885	-	-	-	99

Throughout this thesis, as in Table 1, devices are differentiated by growth. Sample names indicate the material stack while the number indicates the growth. The latter is particularly important in the case of SRO/PZT/YBCO-3. This sample was grown to investigate the influence of the rough YBCO on measurement. Thus, after initial growth of the SRO/PZT/YBCO material stack, the sample was split in half. One half, referred to as SRO/PZT/YBCO-3 was patterned with a YBCO top electrode, while the other half referred to as SRO/PZT/YBCO/Pt-3 was deposited with platinum before patterning. Both SRO/PZT/YBCO-3 and SRO/PZT/YBCO/Pt-3 have the same growth number indicating that common layers were grown together.

3.2 Hysteresis Measurements

Switching dynamics in FeCAPs are commonly studied through hysteresis measurements. These measurements are conducted by applying a triangle, or sine wave, of sufficient amplitude to the electrodes on either side of the ferroelectric and measuring the current

response. During switching, a current flows through the device in the direction of polarization. When the polarization state of the entire ferroelectric layer has switched, it goes back to acting like a typical dielectric. Integrating this current response produces a loop that can be used to extract key information about the FeCAP. This process is shown in Figure 6.

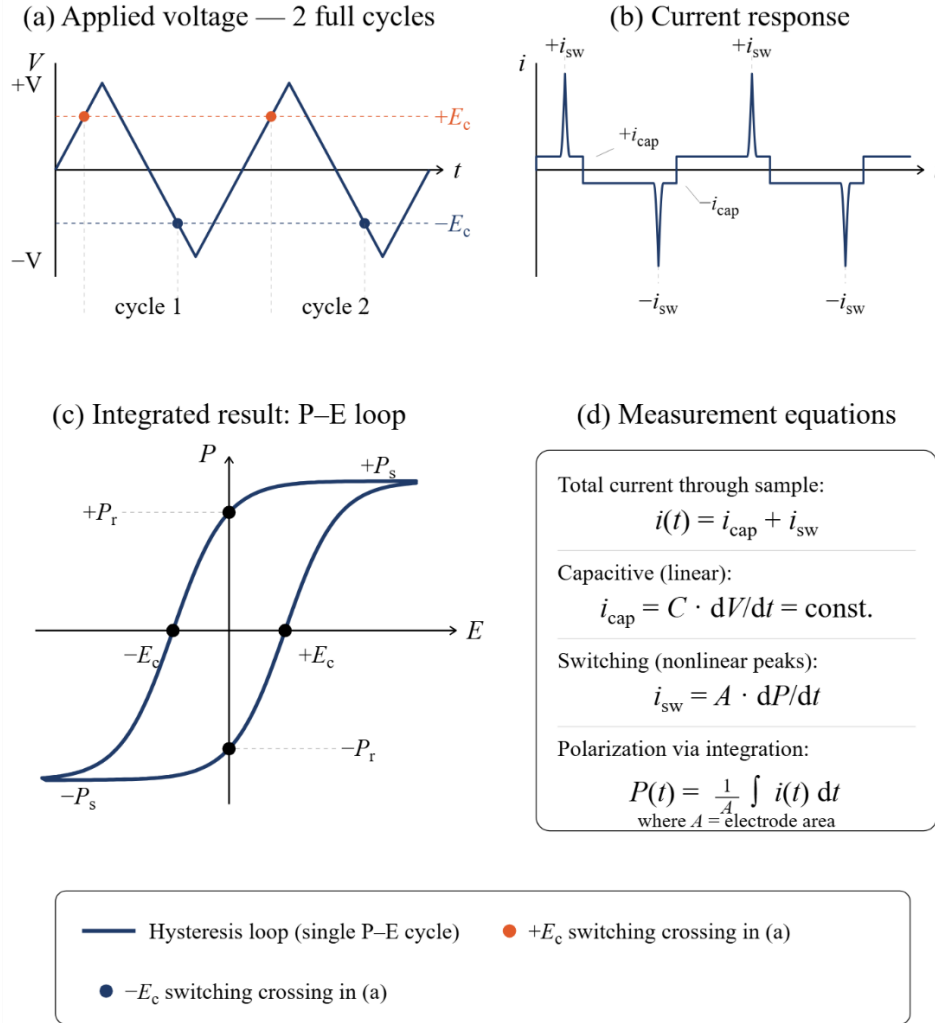


Figure 6 Illustration of a Hysteresis Measurement:

a. Shows the input waveform. **b.** shows the current response from the FeCAP **c.** shows the integrated current response. **d.** Shows the equations used to create the hysteresis loop.

From the hysteresis loop, key performance features of the FeCAP can be extracted. This is shown in Figure 7. Tuning these properties is vital in the design of FeCAPs. For example, in FeRAM it is important to minimize the coercive field (E_c) to minimize power consumption. This must be balanced with maintaining sufficiently high remnant polarization (P_r) so that the capacitor still functions as a memory device.

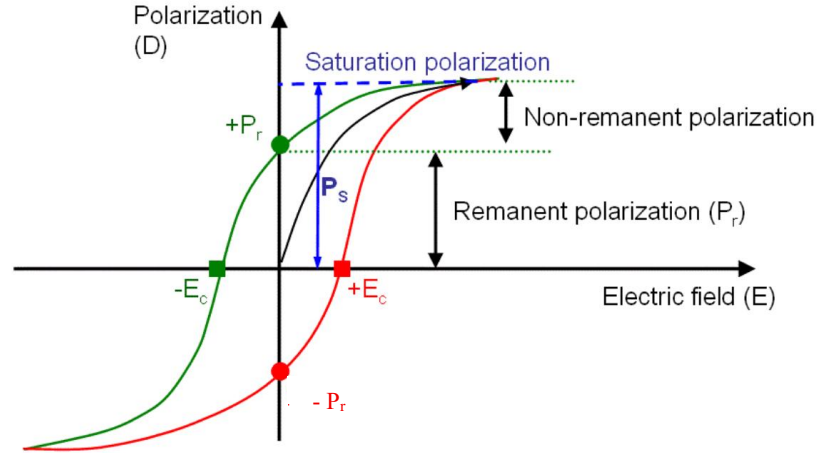


Figure 7 Feature extraction from ferroelectric hysteresis:
From hysteresis loops we can determine the saturation polarization (P_s), remnant polarization (P_r), and coercive field (E_c) of a FeCAP. Figure reprinted from [30].

While hysteresis provides an intuitive way of understanding switching dynamics. It is not a reliable measurement as the loop can be distorted due to nonidealities in the ferroelectric, or measurement system. Famously, a parasitic effect such as leakage can even make non-ferroelectric materials appear otherwise [31]. A summary of the various sources of nonidealities in ferroelectric loops is shown in Figure 8. In real-world measurements a combination of the various nonidealities shown are involved. This makes it particularly difficult to separate parasitic effects from hysteresis.

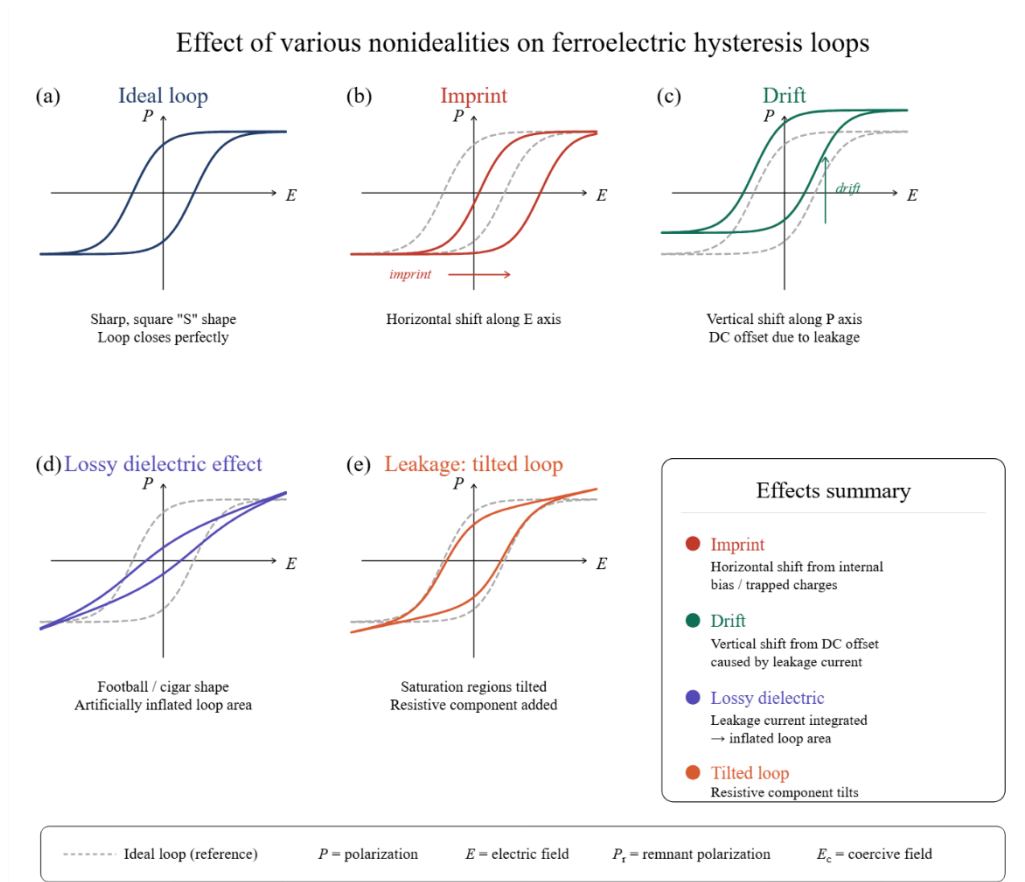


Figure 8 The effects of various nonidealities on hysteresis loops:

a. We see a standard hysteresis loop not affected by nonidealities. **b.** We see the effect of imprint, a horizontal shift resulting from an internal bias field and trapped charges in the ferroelectric. **c.** We see the effect of drift in which the loop moves vertically due to leakage current. **d.** We see the loop for a lossy dielectric, a material that is not ferroelectric but appears to be due to the integration of leakage current. **e.** We see how leakage current adds a resistive component to the FeCAP, tilting the loop.

3.3 PUND Measurements

The unreliability of hysteresis loops prompts the use of a more reliable measurement to quantify ferroelectric switching behavior. PUND, which stands for Positive-Up-Negative-Down, provides a way to do this [32].

We use a three pulse PUND measurement, as seen in Figure 9. The first is the state preparation pulse. It polarizes the FeCAP in the negative state. This is important to ensure that the switching current measured later is for the entire ferroelectric layer, not just a portion. The next pulse, the switching pulse polarizes the FeCAP upwards in the opposite direction. Finally, a third measurement pulse is applied identical to the switching pulse.

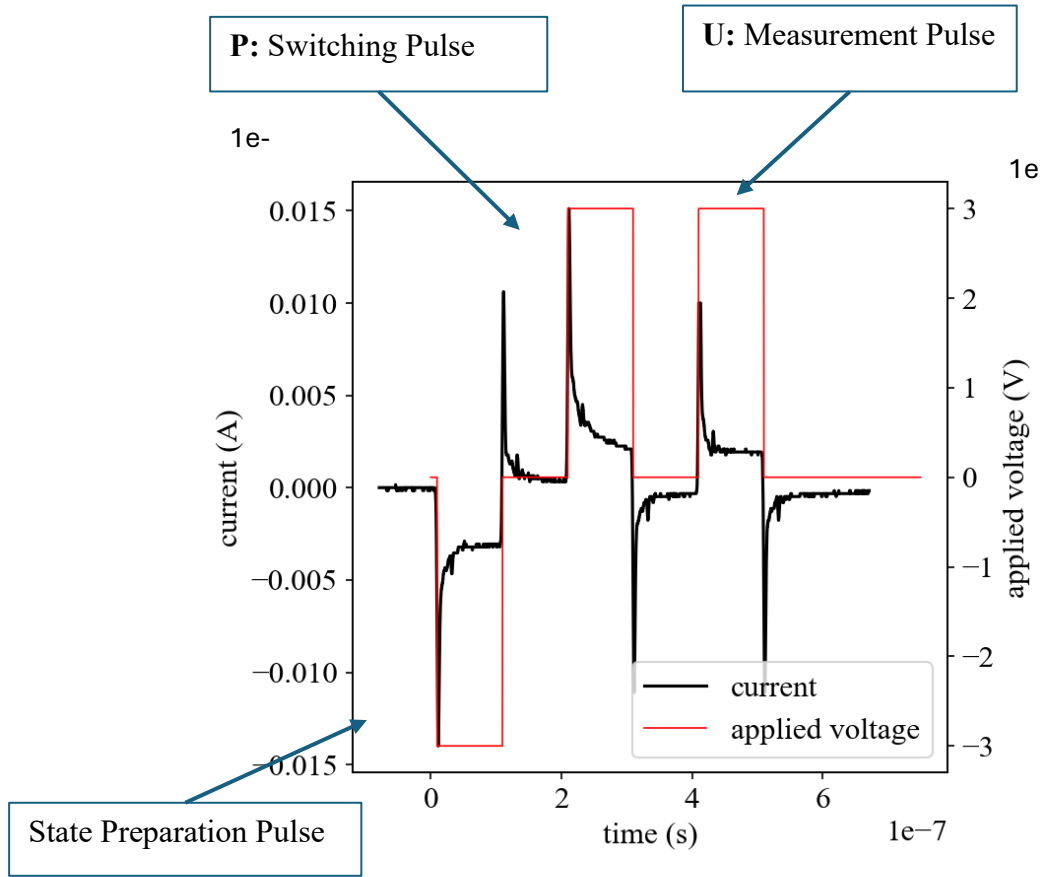


Figure 9 Breakdown of PUND Measurements:

The current response (black) is shown in response to the applied voltage (red). The purpose of a PUND measurement is to isolate the switching current. Thus, to guarantee that switching occurs it is first necessary for a state preparation pulse to polarize the FeCAP downward. The P, switching pulse is then applied to switch the polarization of the ferroelectric layer. The U, measurement pulse follows to measure parasitic effects such as capacitance, and leakage. Notice that the current response for the P pulse is higher than for the U pulse. This is characteristic of ferroelectric switching as the U pulse does not include any switching current.

The total current measured during the switching pulse is given by Equation 3.

$$I_{P \text{ Pulse}} = I_{\text{Switching}} + I_{\text{Dielectric}} + I_{\text{Leakage}} \quad (3)$$

Because all ferroelectric switching occurs in the switching pulse, the total current measured during the measurement pulse is given by Equation 4.

$$I_{U \text{ Pulse}} = I_{\text{Dielectric}} + I_{\text{Leakage}} \quad (4)$$

The switching current, without any nonidealities can be found with Equation 5.

$$I_{\text{switching}} = I_{P \text{ Pulse}} - I_{U \text{ Pulse}} \quad (5)$$

Thus, looking at the behavior of $I_{\text{switching}}$ is much more dependable than hysteresis. Furthermore, integrating this value over the switching pulse and dividing by area gives us P^* , the polarization of the ferroelectric capacitor due only to ferroelectricity.

3.4 Measurement Set Up

Both PUND and hysteresis measurements were conducted on the stochastic FeCAPs. As described above, hysteresis measurements are used to gain a conceptual understanding of switching dynamics while PUND is used to create the PUFs.

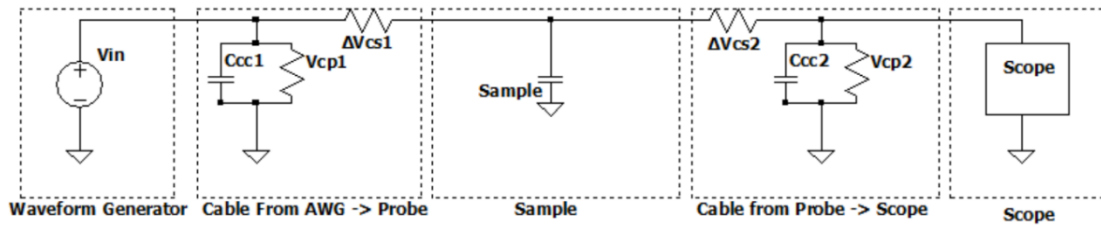


Figure 10 Measurement Circuit:

The measurement circuit consists of two probes, landed on the top and bottom electrodes of the FeCAP respectively. One probe is connected to the output of the waveform generator while the other is connected to the input of an oscilloscope. Once the waveform is applied, the oscilloscope automatically triggers and begins capturing the waveform.

The measurements are completed on a probe station using the circuit shown in Figure 10. Input waveforms are created using a Keysight 81150A Pulse Function Arbitrary Noise Generator. The current response through the FeCAP is read using a Keysight DSOX3024A oscilloscope. Interaction with the FeCAP occurs through coaxial 200 MHz titanium probe tips, which are landed on the top electrode and on silver paint connected to the bottom electrode.

To ensure standardization and reproducibility of measurement, the PIEC Python package [33] was used to automate both the arbitrary waveform generator and the oscilloscope.

For all charge transport measurements, waveforms were applied at approximately 1 MHz frequency, which was the regime of observed stochasticity.

3.5 Data Analysis

All data analysis was conducted within PIEC, which also produced plots for hysteresis and PUND data. The PUF distributions shown later used data produced by PIEC, but were plotted in python using matplotlib [34].

3.6 AFM/PFM

To characterize the quality of thin film growth Atomic Force Microscopy (AFM) is used to assess the surface roughness of thin film depositions. AFM was first theorized by Binnig et al. [35] and has since become a staple of materials characterization. The system works by moving a cantilever with a sharp tip over the surface. A laser is focused on this cantilever and reflected light is read by a photo detector. As the cantilever moves with the high and low spots on the surface, the reflected light read by the photodetector varies. This allows for incredibly precise measurements of surface topography with a resolution on the nanometer scale. The setup is shown in Figure 11.

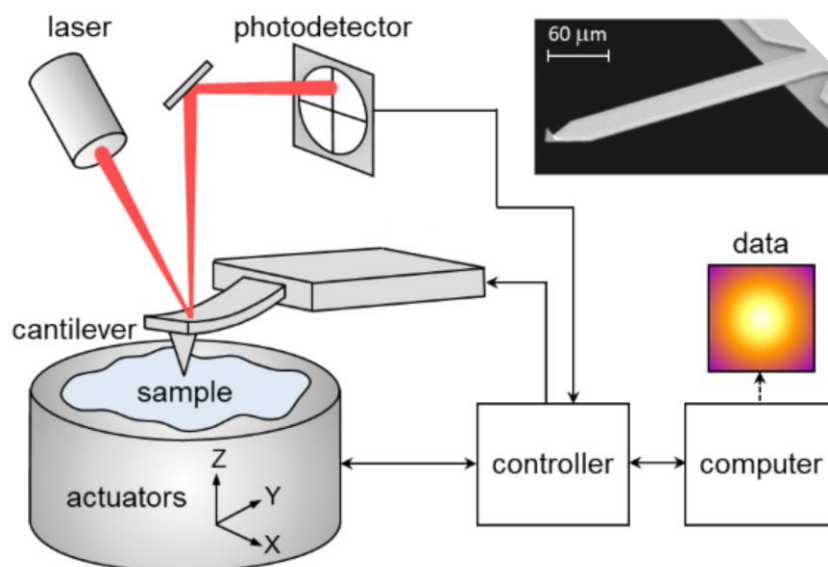


Figure 11 AFM set up
Reprinted from [36].

Piezoresponse Force Microscopy (PFM) is an extension of AFM. In PFM, a voltage is applied to the cantilever tip, in piezoelectric materials this causes a lattice distortion that can be read by the same photodetector mechanism as AFM [37]. This is an incredibly valuable tool for the investigation of ferroelectric domains [38]. With thin oxide electrodes it can also be possible to perform PFM on a ferroelectric layer through the electrode [25].

3.7 XRD

X-Ray Diffraction (XRD) was used to verify the quality of the crystal structure in grown devices. XRD utilizes Bragg's law to measure the difference between reflections in the lattice geometry. These reflections arise from X-Rays directed at the material as different angles are scanned [39]. Constructive interference at certain angles results in peaks when

reflected X-Rays are measured. These peaks are unique to different materials and lattice geometries and allow us to measure crystal structure. An illustration of Bragg's law and the XRD set up are shown in Figure 12 and Figure 13 respectively.

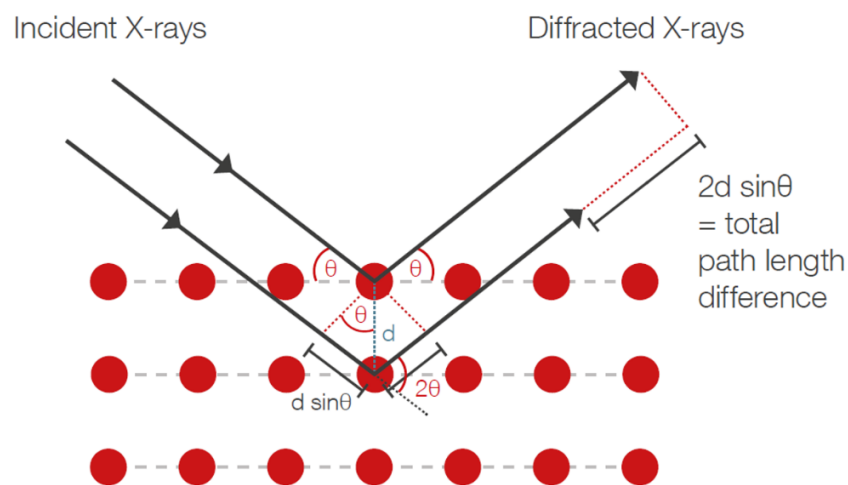


Figure 12 An illustration of Bragg's Law.
Reprinted from [40]

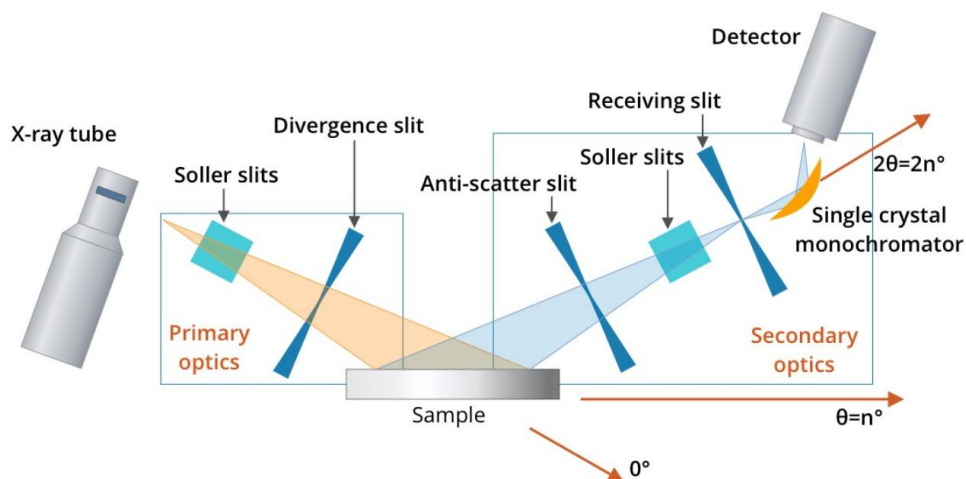


Figure 13 XRD Set Up
Reprinted from [41].

Chapter 4

Results

4.1 Typical Switching Behavior in FeCAPs

Before discussing stochastic results, it is first important to understand typical polarization switching in an FeCAP. Utilizing the Landau-Ginzburg-Devonshire theorem discussed in Chapter 1.2 we can model the material stack discussed above. The results are generated with FeSim and shared in Figure 14. Inherent manufacturing imperfections mean that real devices will not show the exact same behavior. However, it is important to note the orders of magnitude of polarization and current values.

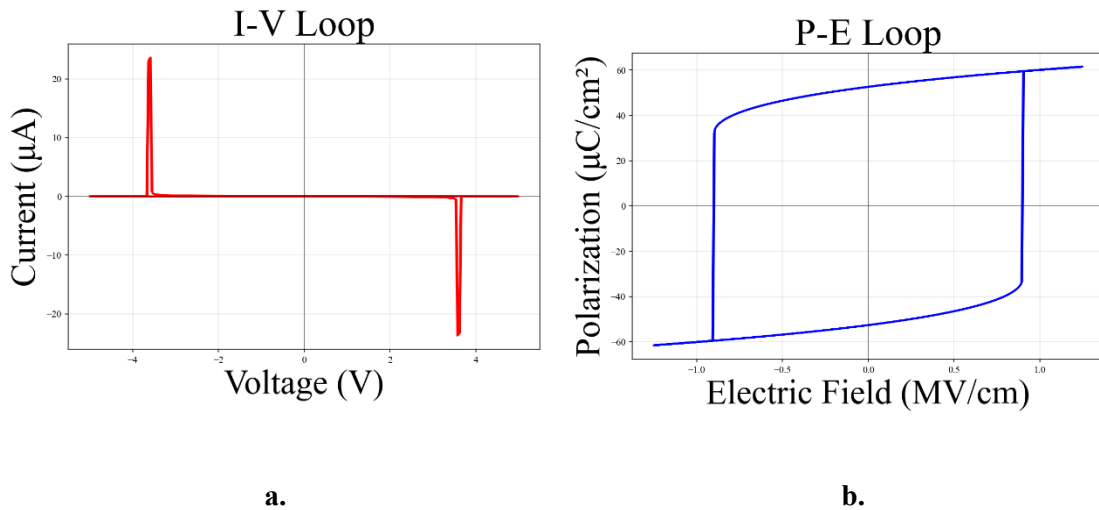


Figure 14 Simulation of Material Stack using LGD:
Of important note are the orders of magnitude of both the current and polarization values. In **a.** we see the simulated current response where current values are on the order of a tenth of a milliamp. In **b.** we see remnant polarization values are around 60 $\mu\text{C}/\text{cm}^2$

In Figure 15 we show experimental results from a known FeCAP with a similar material stack. In this case we measure a SRO/PZT/Pt device that is known to have good ferroelectric response. We see that measured values are approximately modeled values. It should be noted that the experimental loop shows some drift as discussed in Figure 8. This validates the experimental set up and demonstrates that high quality hysteresis data is obtainable with the measurement circuit implemented in this work. Figure 16 further shows the PUND data for the control device. This similarly provides a baseline with which we will compare later. Of note are the distinct single peak current responses to

each applied square wave. Furthermore, the first “P” response is taller than the “U” response, this is how we know the ferroelectric layer has switched. As discussed in Chapter 3.3, the U pulse measurement response does not include switching current and is thus shorter.

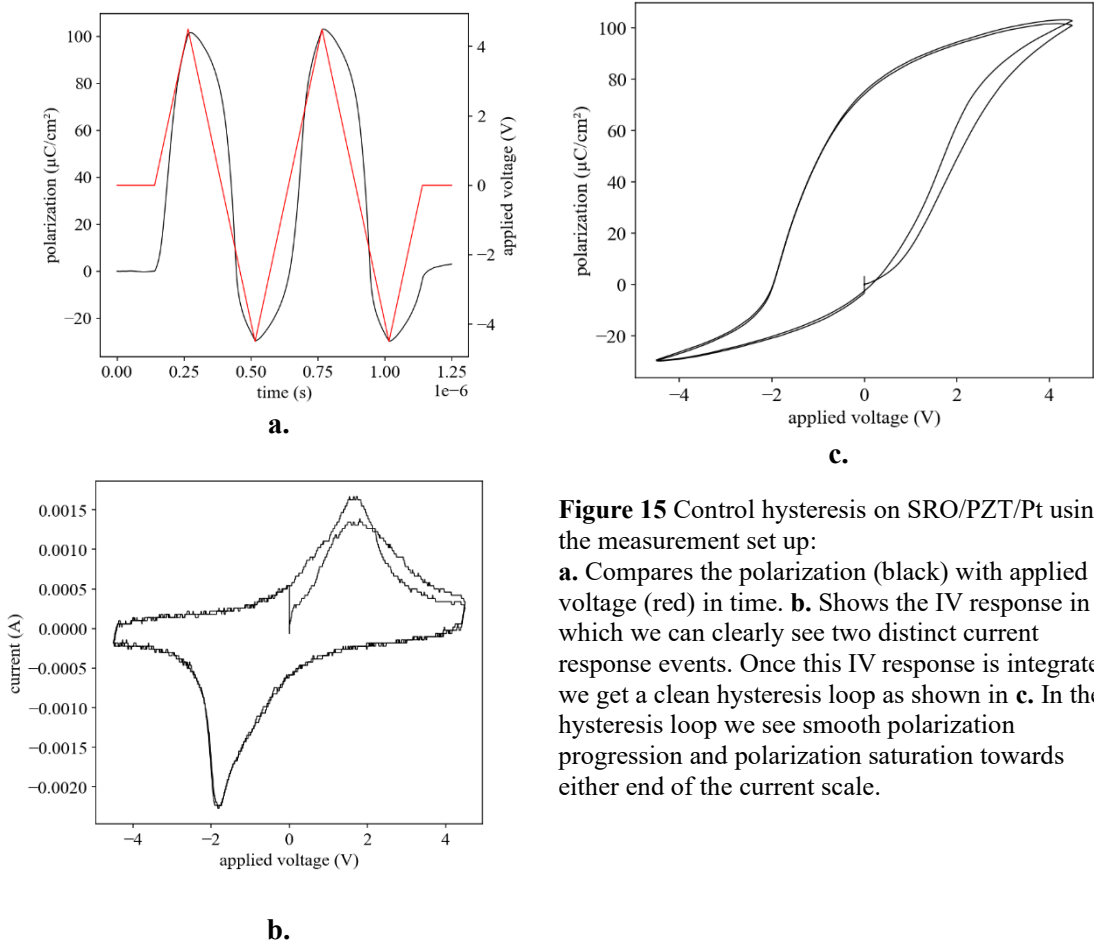


Figure 15 Control hysteresis on SRO/PZT/Pt using the measurement set up:
a. Compares the polarization (black) with applied voltage (red) in time. **b.** Shows the IV response in which we can clearly see two distinct current response events. Once this IV response is integrated, we get a clean hysteresis loop as shown in **c.** In the hysteresis loop we see smooth polarization progression and polarization saturation towards either end of the current scale.

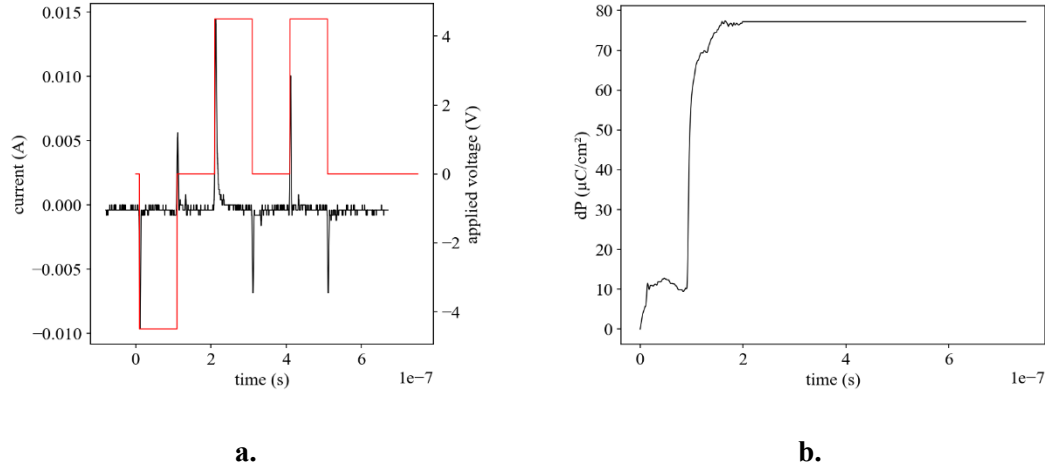


Figure 16 Control PUND on SRO/PZT/Pt using the measurement set up:

Where the black line shows the current response, and red the applied voltage waveform. In **a.** we can see that the current response to the switching pulse is greater than the measurement pulse. This is indicative of ferroelectric switching as the measurement pulse will not include switching current. **b.** shows how the rate of polarization changes over time

4.2 Interrupted Switching Behavior in Stochastic FeCAPs

The Hysteresis data shown for the control device is contrasted with data presented in Figure 17, which shows hysteresis data in stochastic devices. This data is presented across the different stochastic sample growths, and for varying device sizes. Of note in this data are the “step” like features in the hysteresis data. This results from the current spikes seen in the raw measurement trace. These current spikes resemble Barkhausen pulses, the behavior we hope to stimulate in this work. It should be noted that repeating the loop measurements on each device results in different loops shapes (See Figure 18).

In looking at the polarization values achieved in these hysteresis loops we see that SRO/PZT/YBCO-1 is nearly two orders of magnitude smaller than both simulated results and other sample polarizations. This could be due to several factors including the use of an earlier PZT recipe, which had not yet been optimized, dead ferroelectric layers, or other parasitic effects.

As discussed in Chapter 3.3, hysteresis data can be influenced by parasitic effects. Thus, to isolate the stochastic Barkhausen-like charge response, we perform PUND measurements. Examples of these PUND measurements for the same growths and similar diversity of device sizes are shown in Figure 19.

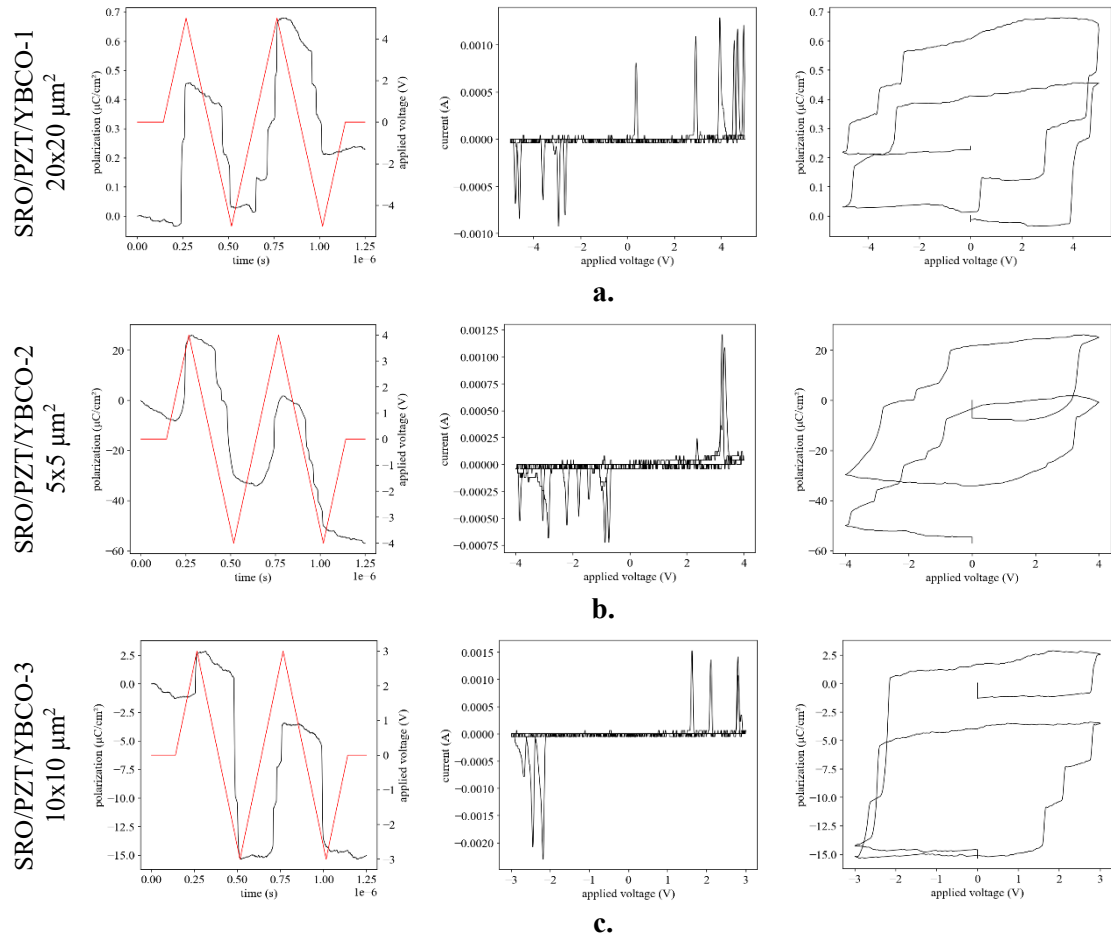


Figure 17 Stochastic Hysteresis Examples Across Growths and Device Size
In this figure we see how current pulses in the IV curves (middle column) show irregular current spikes which translate to the step-like hysteresis behavior. We further see that this occurs across different growths and device sizes.

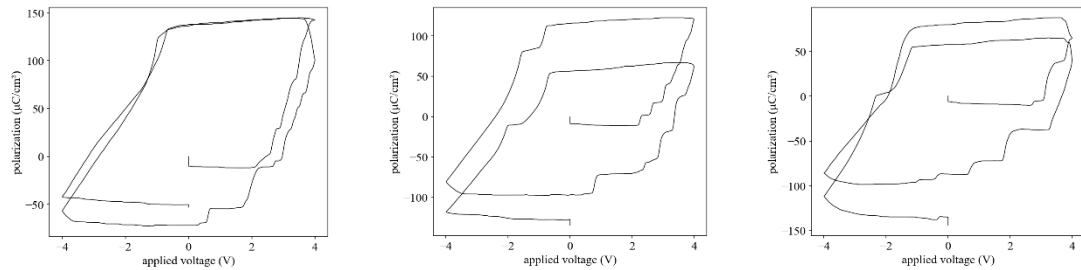


Figure 18 Example Hysteresis Loops for a 10x10 μm^2 device on SRO/PZT/YBCO-2.
In all three cases shown, the same input waveform was applied to the same FeCAP. The differences in hysteresis shape can be attributed to differences in the switching current. These examples demonstrate the variability in switching dynamics seen in the SRO/PZT/YBCO material stack.

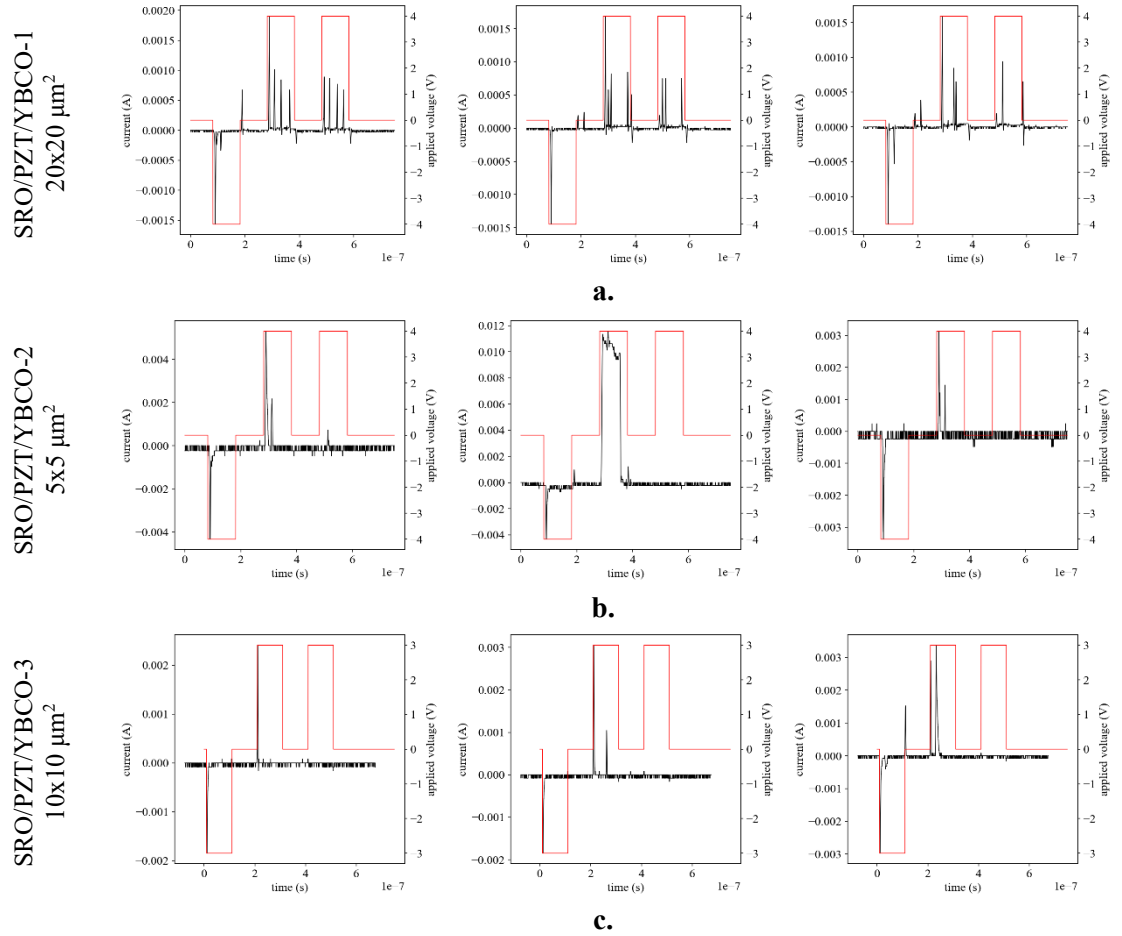


Figure 19 Stochastic PUND Examples Across Growths and Device Size:

In this figure we see examples of PUND responses across growths and device sizes. The current spikes seen are irregular and vary from cycle to cycle. While such variability was seen across growths SRO/PZT/YBCO-1 shown in **a.** had the most variation.

It should be noted that in PUND data we seek to maximize the number of random current spikes that appear. So, while all samples showed some Barkhausen-like activity, the SRO/PZT/YBCO-1 growth showed the best stochastic performance by far.

4.3 Distribution of PUND values

To study a potential distribution in more detail, data from SRO/PZT/YBCO-1 is considered. Despite having polarization values nearly two orders of magnitude below other samples, the SRO/PZT/YBCO-1 growth showed much more cycle-to-cycle stability than other samples. As seen in Figure 20, SRO/PZT/YBCO-1 maintained stochastic performance for at least a thousand cycles while SRO/PZT/YBCO-2 quickly collapsed to zero.

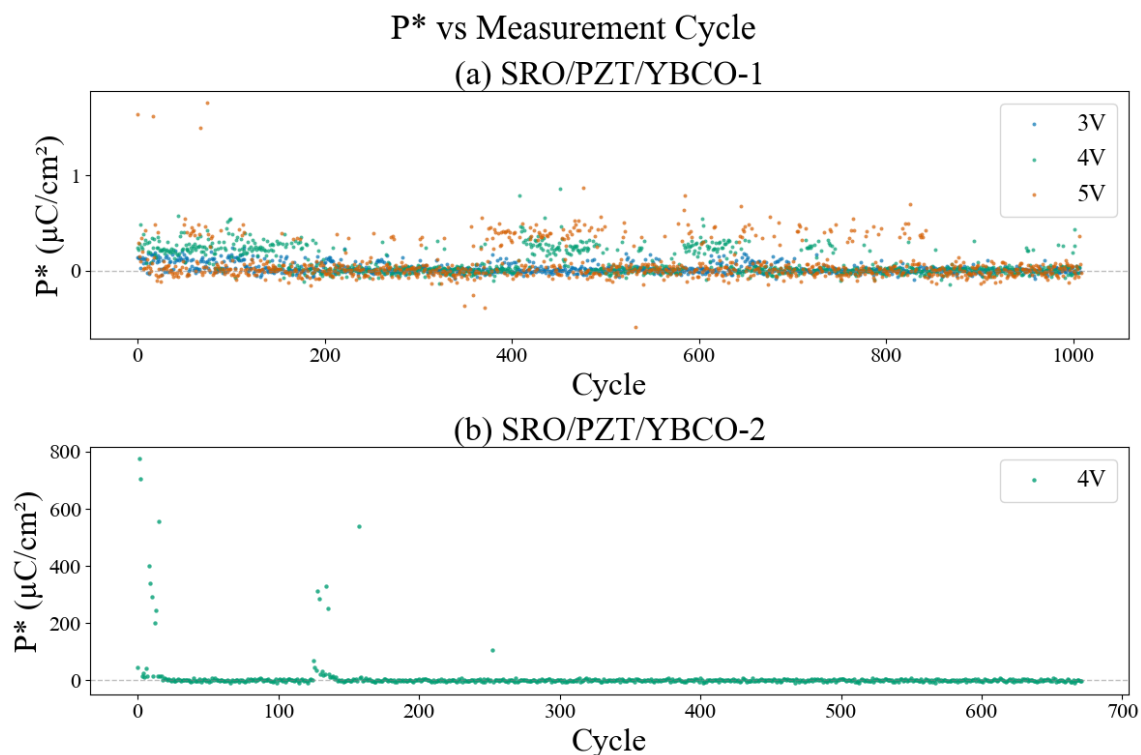


Figure 20 P* v. cycle for **a.** SRO/PZT/YBCO-1 and **b.** SRO/PZT/YBCO-2:

In **a.** we can clearly see that the SRO/PZT/YBCO-1 sample shows variability in P* data for at least 1000 cycles. In **b.** we see that for SRO/PZT/YBCO-2 the polarization values span a broader range, but variability collapses to zero rapidly. This SRO/PZT/YBCO-1 is the more suitable candidate for the creation of a PUF.

The distribution of P* data for SRO/PZT/YBCO-1 is shown in Figure 21. In examining the distribution, we see that the P* values follow a bimodal distribution. At each voltage amplitude there seems to be a spread around zero, and another at a higher value.

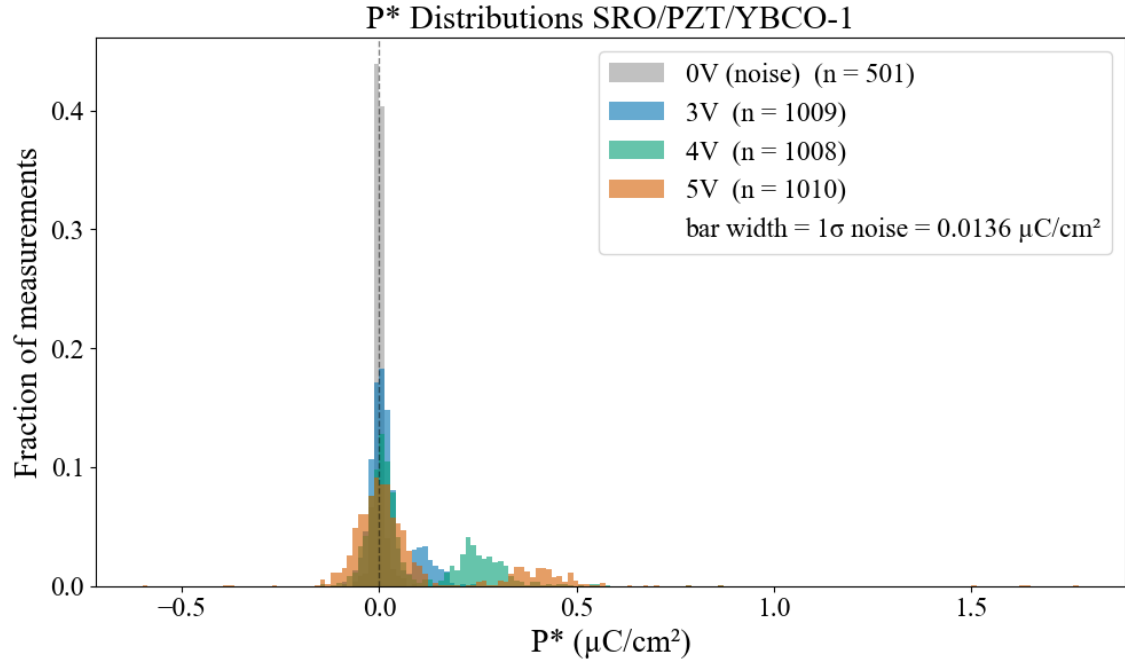


Figure 21 Distribution of P^* values for a $40 \times 40 \mu\text{m}^2$ FeCAP on SRO/PZT/YBCO-1. In this figure, the width of each bar represents one standard deviation of circuit noise. Thus, the distribution of P^* values for all measurement voltages is much wider than circuit noise. Furthermore, we see that distributions follow a bimodal pattern in which one taller peak appears around $0 \mu\text{C}/\text{cm}^2$ and another shorter peak forms between 0 and $0.5 \mu\text{C}/\text{cm}^2$.

The bimodal nature of the distributions is investigated further in Figure 22, where each amplitude distribution is plotted separately. For each amplitude, examples are also shown from both peaks. This allows us to clearly see the dual phenomenon occurring in this sample. At the larger peak, around zero, there appears to be no switching occurring. This is apparent from the relatively flat current trace. Conversely the second smaller peak appears to show ferroelectric switching. This is clearly seen in the large current spikes, where the current trace during the switching pulse is much higher than the current trace during the measurement pulses. In a TRNG device, isolating the second region would be important.

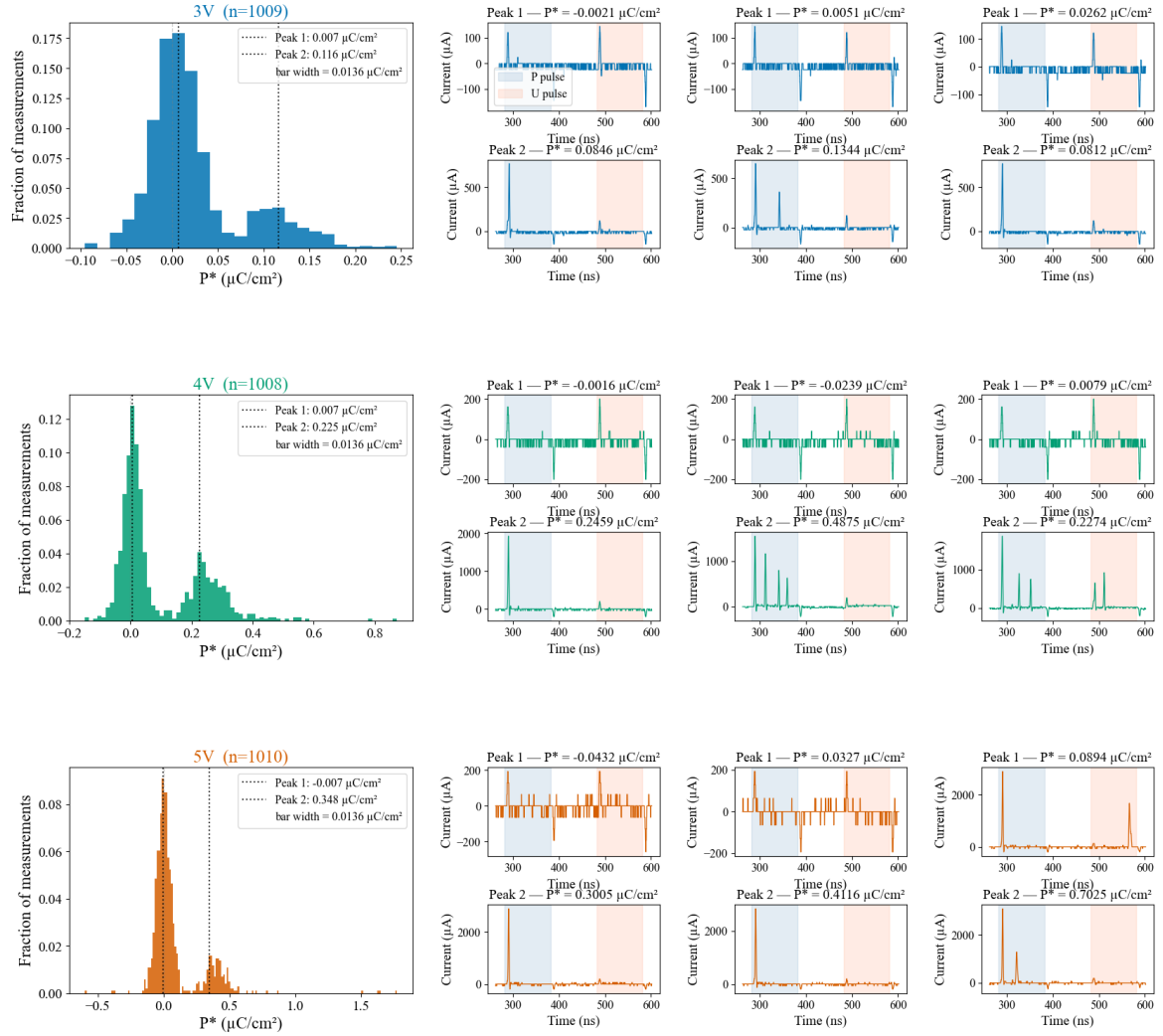


Figure 22 Breakdown of individual PUND distributions for SRO/PZT/YBCO-1:

For each amplitude the distribution is plotted on the left and examples from each peak are shown to the right. The top row shows examples from the first peak while the bottom row shows examples from the second peak. In each case we see that the first peak is largely comprised of nonswitching events, indicated by a relatively flat current trace and no apparent difference in the heights of current response during the P and U pulses. The second peak is comprised of switching events: seen in the taller current response during the P pulse than U pulse. In the second peak we further see variation in current spikes. As amplitude increases the frequency of switching events is seen to increase along with the polarization during such events.

4.4 Materials and Growth Validation

Having determined that a distribution could be made from PUND measurements of the stochastic material stack, it becomes important to investigate circuit level nonidealities that could also explain the stochasticity. As noted in Chapter 3.1, this was done with SRO/PZT/YBCO-3. An AFM image of the unpatterned YBCO layer of

SRO/PZT/YBCO-3 is shown in Figure 23. The average RMS roughness over the 20x20 μm^2 area shown in was $11.56 \text{ nm} \pm 1.077 \text{ nm}$.

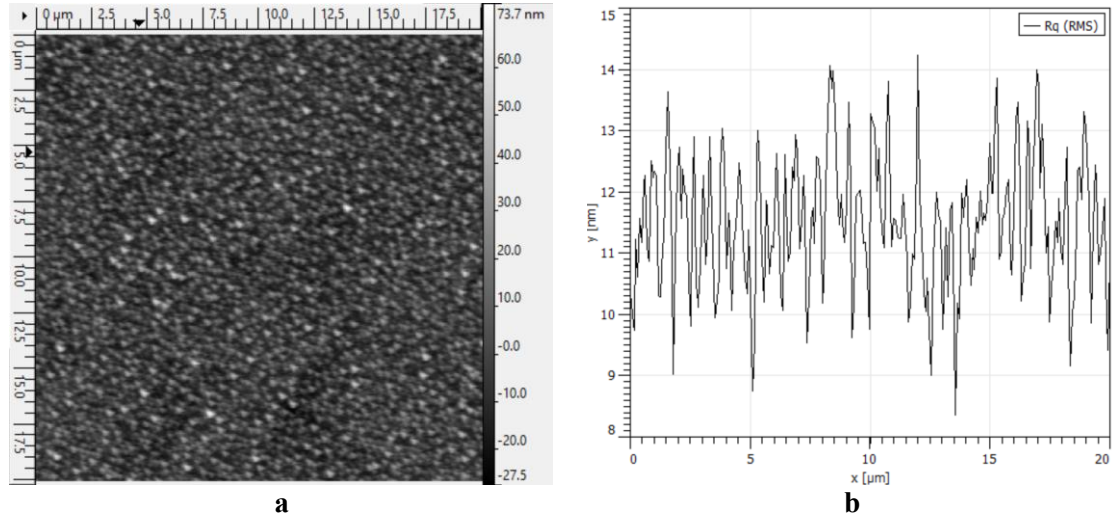


Figure 23 AFM of unpatterned SRO/PZT/YBCO-3:
The average RMS roughness of the 20x20 μm^2 area shown in **a**. is $11.56 \text{ nm} \pm 1.077 \text{ nm}$

AFM was then performed on the patterned SRO/PZT/YBCO-3 sample. This data is presented in Figure 24. The average RMS roughness of the highlighted area is $9.199 \text{ nm} \pm 1.93 \text{ nm}$.

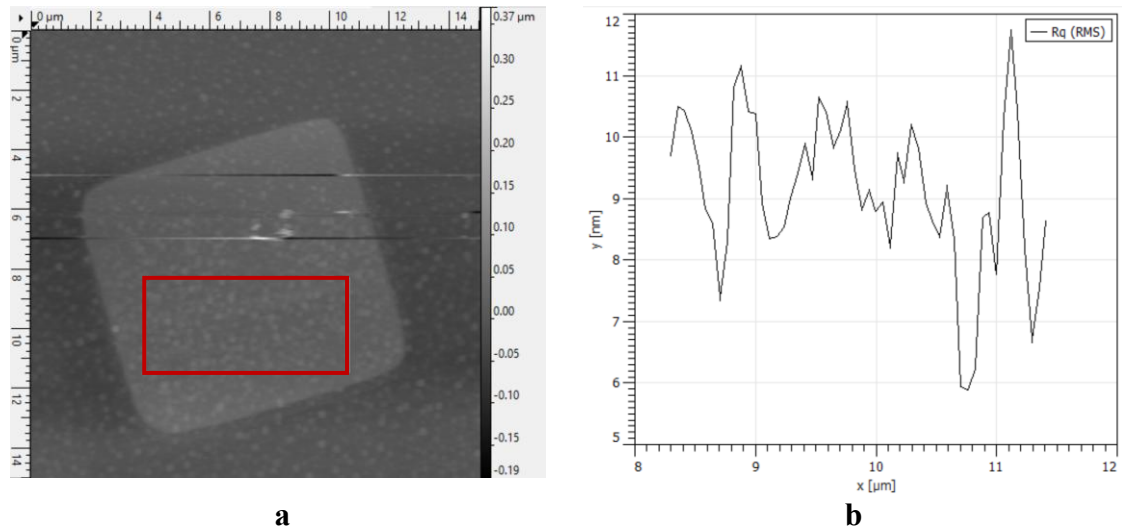


Figure 24 AFM of patterned device on SRO/PZT/YBCO-3:
The average RMS roughness over area shown in **a**. is $9.199 \text{ nm} \pm 1.93 \text{ nm}$.

AFM was also conducted on a device patterned on the same growth with platinum deposited as the top electrode. This is shown in Figure 25. Charge transport data for the SRO/PZT/YBCO/Pt-3 device is shown in Figure 26.

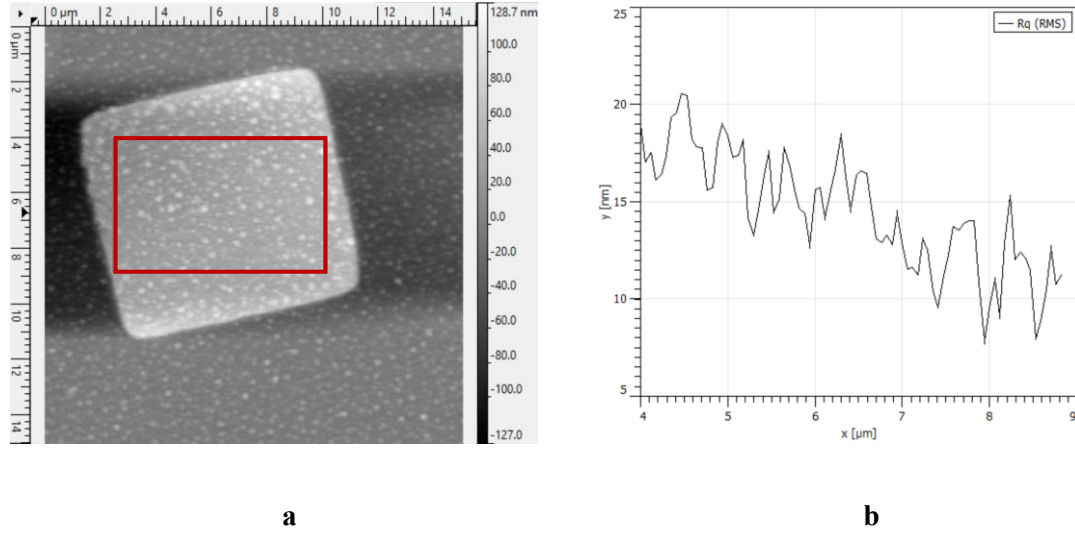


Figure 25 AFM of patterned device on SRO/PZT/YBCO/Pt-3:
The average RMS roughness over area shown in **a.** is $14.810 \text{ nm} \pm 2.901 \text{ nm}$

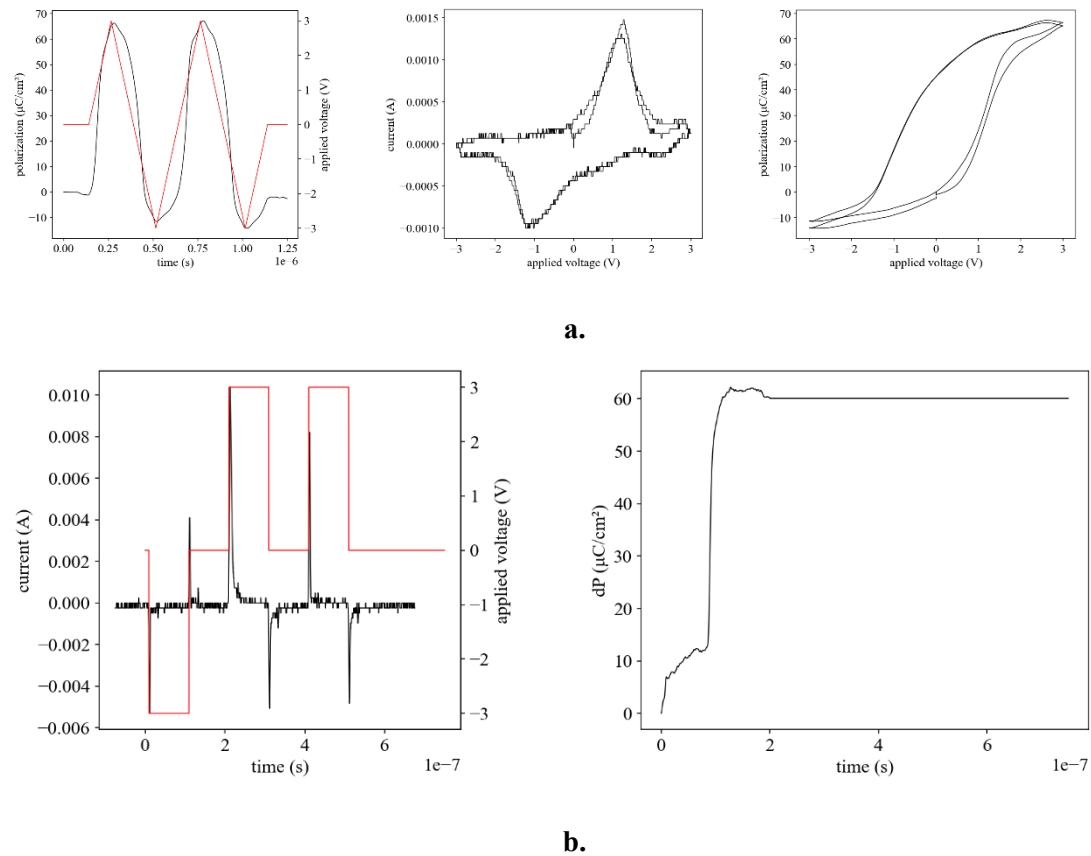


Figure 26 Charge Transport Data taken on SRO/PZT/YBCO/Pt-3:
In **a.** we see hysteresis data in which the loop looks smooth and switching appears continuous. In **b.** we see a PUND measurement without any current spikes. The results of these charge transport measurements demonstrated that the SRO/PZT/YBCO/Pt-3 devices did not have the same stochastic behavior.

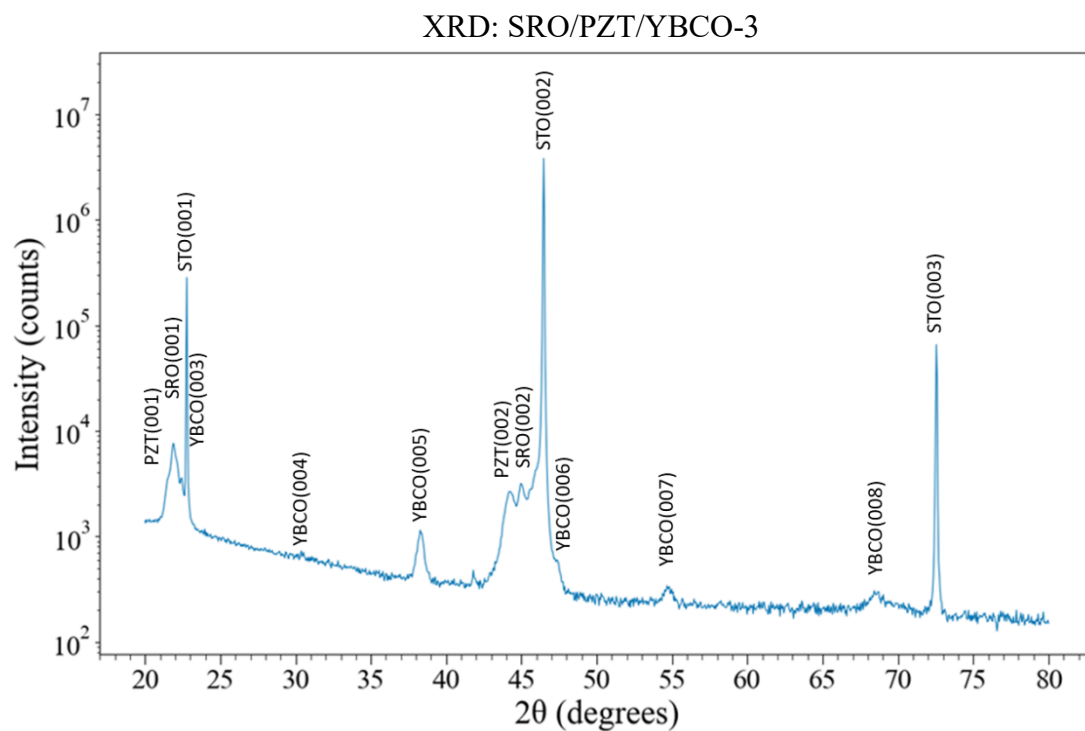


Figure 27 XRD of SRO/PZT/YBCO-3 with labeled peaks:

The plot shows diffraction peaks from XRD of the SRO/PZT/YBCO-3 sample. From the data we confirm growth of the desired thin film layers.

Chapter 5

Discussion and Conclusion

5.1 Interpretation of Charge Transport Measurements

In standard FeCAPs, polarization switching can be measured as a smooth process. A simulation of polarization switching in a ferroelectric of similar thickness and size to the stochastic devices discussed is shown in Figure 14. From this LGD simulation we see that an ideal device remnant polarization is on the order of $60 \mu\text{C}/\text{cm}^2$. We compare the ideal case to a control device composed of a SRO/PZT/Pt material stack, whose data is presented in Figure 15 and Figure 16. Despite some drift, we see that the control device achieves a similar remnant polarization of $\sim 50 \mu\text{C}/\text{cm}^2$ (Figure 15.c). The charge transport measurements of the control device crucially serve to validate the measurement set up, demonstrating measurements comparable to the ideal case.

In this work, we attempted to design a FeCAP in which random nucleation and domain growth during polarization switching can be used to create a physically unclonable function for TRNG. The basis of this PUF is the current measured during the switching process. In normal hysteresis and PUND data (shown in Figure 15 and Figure 16), the measured current through the FeCAP does not show much change from cycle to cycle. By contrast, our design hoped to promote less stable behavior. Through the introduction of defects at the ferroelectric-electrode interface, we aim to create pinning sites for polarization switching. This pinning could then create measurable cycle-to cycle differences in switching current. Measurements of all three stochastic samples showed this desired behavior in hysteresis (see Figure 17 and Figure 18). When contrasted with Figure 15, we see that the hysteresis loops of stochastic samples are less smooth and have a step-like behavior. These steps resemble those of Barkhausen pulses, which arise when switching domains overcome pinning sites causing a large volume of material to switch quickly. This behavior is further seen in the current spikes shown in graphs of applied voltage v . measured current. Furthermore, Figure 18 shows how this hysteresis behavior changes from cycle to cycle. As discussed, this is a desired behavior. It should be noted that SRO/PZT/YBCO-1 shown in Figure 17.a has polarization values roughly two orders of magnitude smaller than other growths. This could be attributed to several factors stemming from a nonidealized PZT recipe that was used to create this earliest device. Later devices (shown in Figure 17.b-c) were produced with an optimized PZT recipe and show polarization values on the same order of magnitude as the simulated hysteresis shown in Figure 14.

When examining the PUND data for the stochastic samples, stochasticity can again be observed in the differences in cycle-to-cycle current responses as shown in Figure 19. This is especially important as PUND measurements form the basis of the PUF discussed later. It should be noted that all stochastic growths show this cyclic variation.

5.2 Creating a PUF from PUND data

As discussed previously P^* was selected as the metric to create the PUF as it isolates switching behavior. Although the SRO/PZT/YBCO-1 growth showed the lowest polarization values across the stochastic devices produced, it was selected to create the PUF. This was due to the SRO/PZT/YBCO-1 growth's stability over several cycles. A comparison of SRO/PZT/YBCO-1 at various amplitudes with SRO/PZT/YBCO-2 is shown in Figure 20. Variation in P^* for SRO/PZT/YBCO-1 is maintained for hundreds of cycles while SRO/PZT/YBCO-2 quickly collapses to zero. SRO/PZT/YBCO-3 showed similar behavior to SRO/PZT/YBCO-2.

A distribution of the P^* data for SRO/PZT/YBCO-1 is shown in Figure 21 where the width of a single bar in the histogram represents one standard deviation of the circuit noise. In the figure we see that the P^* distribution of SRO/PZT/YBCO-1 is much wider than circuit noise and thus creates a relevant distribution. We further see that the distributions appear to be bimodal.

The bimodal distributions are further investigated in Figure 22, where distributions shown in Figure 21 are plotted individually alongside example current traces from both peaks. From this, we see that the first taller peak can be attributed to a non-switching event. This is seen in the similar heights of current response between the P and U pulses in the PUND, suggesting switching current was near zero. Conversely, the second peak appears to be due to switching events. This is characteristically seen in the much larger current response during the P pulse when compared to the U pulse. As amplitude is increased the share of measurements in the first peak is seen to decrease, meaning the number of switching events increases.

5.3 Other Sources of Stochasticity

To confirm the origins of stochasticity, extensive material characterization of the stochastic devices was necessary. The SRO/PZT/YBCO-3 growth was conducted specifically for this purpose, and all charge transport measurements were conducted after material characterization of the sample. X-Ray Diffraction, shown in Figure 27, first confirmed the material layers were deposited correctly.

An attempt was then made to conduct PFM of the ferroelectric layer through the YBCO oxide, as shown in [25]. However, PFM through the oxide was not possible due to the rough YBCO surface (see Figure 23).

To investigate if this roughness had an effect on the electrical contact, and thus stochasticity, the SRO/PZT/YBCO-3 sample was split in half. As outlined in Chapter 3.1, one half was kept as is and patterned with the typical YBCO top electrode while the other half was deposited with platinum before patterning. The latter sample is henceforth referred to as SRO/PZT/YBCO/Pt-3. The AFM of both SRO/PZT/YBCO-3 (Figure 24) and SRO/PZT/YBCO/Pt-3 (Figure 25) show average RMS roughness of $9.199 \text{ nm} \pm 1.93 \text{ nm}$ and $14.810 \text{ nm} \pm 2.901 \text{ nm}$ respectively. Because these values are comparable, we can investigate the role of electrode material.

We have already established that the SRO/PZT/YBCO-3 samples show stochastic features (Figure 17 and Figure 19). In Figure 26 we show that the SRO/PZT/YBCO/Pt-3 does not. Thus, we see that the addition of a platinum layer removes stochastic behavior. This is likely due to the interaction of the probe tip with the electrode material. When the probe tip lands on the hard ceramic YBCO, natural environmental vibrations could cause the tip to bump into high and low spots on the electrode. This intermittent contact could be the cause of the current pulses in both hysteresis and PUND. Conversely, when the probe tip lands on the softer platinum, it can embed itself, limiting the effects of vibration and establishing strong electrical contact. This is likely why its addition of the SRO/PZT/YBCO-3 growth eliminated stochastic effects. It can thus be concluded that stochastic behavior likely resulted from the interaction of the probe tip with the hard and rough YBCO electrode surface and not pinning during polarization.

5.4 Limitations

Ultimately the observed stochasticity could not be attributed to the pinning of propagating domains as intended. This was limited by the quality of the YBCO growth. The rough surface did not allow for PFM imaging of the ferroelectric layer, and likely caused the stochastic effects observed. High quality epitaxial growth of YBCO has been demonstrated [26] and should first be replicated in any future work on the design proposed in this thesis. Furthermore, such work should utilize platinum as a standard for electrodes, as they ensure complete contact with the measurement probes.

Furthermore, data from Figure 21 suggests that stochastic events are positively correlated with measurement amplitude. The 10 V_{pp} limit on the waveform generator used in our measurement set up limited further study of this correlation at the desired 1 MHz frequency. Future work could use more advanced equipment to further study this trend.

5.5 Conclusion and Future Work

In this thesis we explore a novel approach to True Random Number Generation (TRNG) using thin film ferroelectric capacitors (FeCAPs). Unlike previous approaches to ferroelectric TRNG, we suggest creating probability distributions from switching dynamics, rather than the end state of polarization switching. To accomplish this, we propose a FeCAP with an SRO/PZT/YBCO material stack. The YBCO layer was particularly chosen because of its inherent oxygen deficient growth, which introduces oxygen vacancies at the interface. These oxygen vacancies could act as pinning sites, amplifying the measurable effects of the already probabilistic polarization switching process.

Through charge transport measurement, we show cycle-to-cycle variation in both hysteresis and PUND indicative of the Barkhausen-like effect we hope to stimulate. Using PUND measurements, we further create a distribution of P^* values demonstrating significant distribution widths when compared to circuit noise. This successfully demonstrated that variations in switching currents can be used to create probability distributions.

Material characterization of the FeCAP revealed that stochasticity was likely the result of contact issues between the YBCO electrode and probe tip and not the desired interfacial effects. While the implementation used in this work did not prove successful, the creation of a wide probability distribution from changes in switching dynamics was achieved. Furthermore, the idea of using pinning sites to create interrupted switching has not yet been disproved. Future work in optimization of material growth and defect engineering should certainly be pursued. If variation in switching current can be derived from such efforts, ferroelectric switching dynamics provide an incredibly promising avenue for TRNG.

References

- [1] L. Coppolino, S. D'Antonio, G. Mazzeo, and L. Romano, "A comprehensive survey of hardware-assisted security: From the edge to the cloud," *Internet of Things*, vol. 6, 2019, doi: 10.1016/j.iot.2019.100055.
- [2] U. Rührmair, F. Sehnke, J. Sölter, G. Dror, S. Devadas, and J. Schmidhuber, "Modeling attacks on physical unclonable functions," presented at the Proceedings of the 17th ACM conference on Computer and communications security, Chicago, Illinois, USA, 2010. [Online]. Available: <https://doi.org/10.1145/1866307.1866335>.
- [3] Y. C. Chien, H. Xiang, J. Wang, Y. Shi, X. Fong, and K. W. Ang, "Attack Resilient True Random Number Generators Using Ferroelectric-Enhanced Stochasticity in 2D Transistor," *Small*, vol. 19, no. 38, p. e2302842, Sep 2023, doi: 10.1002/sml.202302842.
- [4] S. Chatterjee, N. Rangarajan, S. Patnaik, D. Rajasekharan, O. Sinanoglu, and Y. S. Chauhan, "FerroCoin: Ferroelectric Tunnel Junction-Based True Random Number Generator," *IEEE Transactions on Emerging Topics in Computing*, vol. 11, no. 2, pp. 541-547, 2023, doi: 10.1109/tetc.2022.3210401.
- [5] M. E. Lines and A. M. Glass, *Principles and Applications of Ferroelectrics and Related Materials*. New York: Oxford University Press, 1977.
- [6] L. Li, L. Xie, and X. Pan, "Real-time studies of ferroelectric domain switching: A review," *Reports on Progress in Physics*, vol. 82, 11/07 2019, doi: 10.1088/1361-6633/ab28de.
- [7] P. Chandra and P. B. Littlewood, "A Landau Primer for Ferroelectrics," 2006.
- [8] N. A. Pertsev, A. G. Zembilgotov, and A. K. Tagantsev, "Effect of Mechanical Boundary Conditions on Phase Diagrams of Epitaxial Ferroelectric Thin Films," *PHYSICAL REVIEW LETTERS*, vol. 80, 9, 1998.
- [9] R. Pankaj, "FeSim: AN OPEN-SOURCE SOFTWARE FOR FERROELECTRIC HYSTERESIS MODELING AND ANALYSIS," Department of Electrical Engineering, Brown University, Providence, Rhode Island, 2026.
- [10] A. Sheikholeslami and P. G. Gulak, "A survey of circuit innovations in ferroelectric random-access memories," *Proceedings of the IEEE*, vol. 88, no. 5, pp. 667-689, 2000, doi: 10.1109/5.849164.
- [11] Y. Ishibashi and Y. Takagi, "Note on Ferroelectric Domain Switching," *Journal of the Physical Society of Japan*, vol. 31, 2, 1971.
- [12] T. Tybell, P. Paruch, T. Giamarchi, and J.-M. Triscone, "Domain wall creep in epitaxial ferroelectric Pb(Zr_{0.2}Ti_{0.8})O₃ thin films," *Physical Review Letters*, vol. 89, 2002, doi: <https://doi.org/10.1103/PhysRevLett.89.097601>.
- [13] Y. H. Shin, I. Grinberg, I. W. Chen, and A. M. Rappe, "Nucleation and growth mechanism of ferroelectric domain-wall motion," *Nature*, vol. 449, no. 7164, pp. 881-4, Oct 18 2007, doi: 10.1038/nature06165.
- [14] W. J. Merz, "Domain Formation and Domain Wall Motions in Ferroelectric BaTiO₃ Single Crystals," *Physical Review*, vol. 95, no. 3, pp. 690-698, 1954, doi: 10.1103/PhysRev.95.690.
- [15] R. Landauer, "Electrostatic Considerations in BaTiO₃ Domain Formation during Polarization Reversal," *Journal of Applied Physics*, vol. 28, no. 2, pp. 227-234, 1957, doi: 10.1063/1.1722712.

- [16] J. Y. Jo *et al.*, "Nonlinear dynamics of domain-wall propagation in epitaxial ferroelectric thin films," *Phys Rev Lett*, vol. 102, no. 4, p. 045701, Jan 30 2009, doi: 10.1103/PhysRevLett.102.045701.
- [17] M. G. Han *et al.*, "Interface-induced nonswitchable domains in ferroelectric thin films," *Nat Commun*, vol. 5, p. 4693, Aug 18 2014, doi: 10.1038/ncomms5693.
- [18] C. D. Tan, J. Gardner, F. D. Morrison, E. K. H. Salje, and J. F. Scott, "Studies of Barkhausen Pulses in Ferroelectrics," *arXiv*, 2018.
- [19] H. Mulaosmanovic, T. Mikolajick, and S. Slesazeck, "Random Number Generation Based on Ferroelectric Switching," *IEEE Electron Device Letters*, vol. 39, no. 1, pp. 135-138, 2018, doi: 10.1109/led.2017.2771818.
- [20] (2010). *NIST Special Publication 800-22 Revision 1a: A Statistical Test Suite for Random and Pseudorandom Number Generators for Cryptographic Applications*. [Online] Available: <https://csrc.nist.gov/external/nvlpubs.nist.gov/nistpubs/Legacy/SP/nistspecialpublication800-22r1a.pdf>
- [21] T. Li *et al.*, "Demonstration of high-reconfigurability and low-power strong physical unclonable function empowered by FeFET cycle-to-cycle variation and charge-domain computing," *Nat Commun*, vol. 16, no. 1, p. 189, Jan 2 2025, doi: 10.1038/s41467-024-55380-x.
- [22] H. Tang *et al.*, "CMOS-Compatible LiNbO₃ Domain-Wall Entropy Engine for Ultrafast True Random Number Generation," *Adv Mater*, vol. 38, no. 4, p. e10981, Jan 2026, doi: 10.1002/adma.202510981.
- [23] S. Kang *et al.*, "Ferroelectric Stochasticity in 2D CuInP(2)S(6) and Its Application for True Random Number Generator," *Adv Mater*, vol. 37, no. 26, p. e2406850, Jul 2025, doi: 10.1002/adma.202406850.
- [24] M. Ramesh, A. Verma, and A. Ajoy, "Proposal for True Random Number Generation in ferroelectric capacitors using noise-induced switching," presented at the 2024 8th IEEE Electron Devices Technology & Manufacturing Conference (EDTM), 2024.
- [25] A. Gruverman, D. Wu, and J. F. Scott, "Piezoresponse force microscopy studies of switching behavior of ferroelectric capacitors on a 100-ns time scale," *Phys Rev Lett*, vol. 100, no. 9, p. 097601, Mar 7 2008, doi: 10.1103/PhysRevLett.100.097601.
- [26] R. Ramesh *et al.*, "Fatigue and retention in ferroelectric Y-Ba-Cu-O/Pb-Zr-Ti-O/Y-Ba-Cu-O heterostructures," *Applied Physics Letters*, vol. 61, pp. 1537-1539, 1992.
- [27] Y. Noguchi, H. Matsuo, Y. Kitanaka, and M. Miyayama, "Ferroelectrics with a controlled oxygen-vacancy distribution by design," *Sci Rep*, vol. 9, no. 1, p. 4225, Mar 12 2019, doi: 10.1038/s41598-019-40717-0.
- [28] U. Petralanda, M. Kruse, H. Simons, and T. Olsen, "Oxygen Vacancies Nucleate Charged Domain Walls in Ferroelectrics," *Phys Rev Lett*, vol. 127, no. 11, p. 117601, Sep 10 2021, doi: 10.1103/PhysRevLett.127.117601.
- [29] D. Dijkkamp *et al.*, "Preparation of Y-Ba-Cu oxide superconductor thin films using pulsed laser evaporation from high T_c bulk material," *Applied Physics Letters*, vol. 51, pp. 619-621, 1987.
- [30] Y. Liao, *Practical electron microscopy and database*. 2006.
- [31] J. F. Scott, "Ferroelectrics go bananas," *Journal of Physics: Condensed Matter*, vol. 20, no. 2, 2008, doi: 10.1088/0953-8984/20/02/021001.

- [32] Z. Gao, S. Lyu, and H. Lyu, "Frequency dependence on polarization switching measurement in ferroelectric capacitors," *Journal of Semiconductors*, vol. 43, no. 1, 2022, doi: 10.1088/1674-4926/43/1/014102.
- [33] *PIEC: Python Integrated Experimental Control*. (2026). GitHub. [Online]. Available: <https://github.com/ELPsyKurusu/piec>
- [34] J. D. Hunter, "Matplotlib: A 2D Graphics Environment," *Computing in Science & Engineering*, vol. 9, no. 3, pp. 90-95, 2007, doi: 10.1109/MCSE.2007.55.
- [35] G. Binnig, C. F. Quate, and C. Gerber, "Atomic force microscope," *Phys Rev Lett*, vol. 56, no. 9, pp. 930-933, Mar 3 1986, doi: 10.1103/PhysRevLett.56.930.
- [36] A. Research. "What is Atomic Force Microscopy (AFM)?" Oxford Instruments. <https://afm.oxinst.com/outreach/atomic-force-microscopy> (accessed).
- [37] A Gruverman, O Auciello, R Ramesh, and H. Tokumoto, "Scanning force microscopy of domain structure in ferroelectric thin films: imaging and control," *Nanotechnology*, 1997.
- [38] Stephen Jesse, Arthur P. Baddorf, and S. V. Kalinin, "Switching spectroscopy piezoresponse force microscopy of ferroelectric materials.pdf," *Applied Physics Letters*, vol. 88, 2006, doi: 10.1063/1.2172216.
- [39] BD Cullity and S. Stock, *Elements of X-Ray Diffraction*. Edinburgh: Pearson, 2014.
- [40] A. Paar. "X-ray Diffraction (XRD)." <https://wiki.anton-paar.com/en/x-ray-diffraction-xrd/> (accessed).
- [41] M. Australia, "X-ray Diffraction," *MyScope Microscopy Training*.